From a data-driven materials ranking to microstructure: thermal-shock damage and creep of BOF refractory linings simulated with calibrated Kanapy RVEs, three-dimensional Abaqus models and machine-learning surrogates

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HIGHLIGHTS

- DFT-based ML and TOPSIS ranking of 66 BOF refractories identifies thermal-shock damage as the deciding criterion
- Periodic Kanapy RVEs, energy-consistent sequentially linear analysis and Gauss-point Norton creep
- Five parameters calibrated on open brick data; graphite effect on stiffness and strength predicted within −16 and −6 %
- Independent Abaqus solution of 60 RVEs agrees in stiffness, creep and post-shock stiffness
- 3-D cohesive RVE: MgO–C keeps 12 % of the elastic hot-face stress after 400 K; creep 3.5× below the 2-D estimate

ABSTRACT

Refractory linings of basic oxygen furnaces (BOF) are selected with index-based rankings that reduce a porous multiphase brick to single numbers such as the Hasselman parameters. A data-driven ranking of 66 refractories, built on machine-learning models trained on density-functional-theory data and the multi-criteria method TOPSIS, shows that the choice between MgO–ZrO_2, magnesia and MgO–C depends almost entirely on how thermal-shock damage is weighted: MgO–C, the industrial standard, ranks first only when the weight of thermal-shock resistance reaches 0.28. To resolve this criterion physically we develop a microstructure-resolved model of thermal-shock cracking and high-temperature creep. Periodic two-dimensional representative volume elements (1 mm, 64 × 64 pixels) are generated from Kanapy ellipsoid packings for four lining families and sampled by a Latin-hypercube design of experiments (320 microstructures). Constrained cooling of the hot face is solved by sequentially linear analysis with an energy-consistent saw-tooth law, and compressive creep by a Norton law integrated at the Gauss points. Five parameters are calibrated on published data of one MgO–C brick and one magnesia brick; without refitting, the model predicts the stiffness and strength ratio of a second MgO–C brick within −16 and −6 % and magnesia creep at 1500 °C within 0.01–0.22 dex at 1.5–2.5 MPa. An independent solution of 60 calibrated models with Abaqus/Standard gives identical moduli, creep strains within 0.5 % and post-shock stiffness within 0.016 up to 100 K. MgO–C initiates cracks first (median 26 K) but retains 0.87–0.92 of its stiffness after 100 K, against 0.55–0.56 for MgO–ZrO_2 and magnesia, and creeps 5–19 times less. As an independent check of the top-ranked carbon grade, three-dimensional Abaqus models with zero-thickness cohesive elements on all 13 645 grain-boundary, graphite and binder interfaces confirm the two-dimensional mechanism and quantify its plane-stress bias: the hot face forms a distributed microcrack network and still carries 12 % of the elastic stress after 400 K, and the connected graphite skeleton reduces creep 3.5-fold relative to the two-dimensional estimate. Brick-scale XFEM and contour-integral models give a hot-face crack onset of 72 K and stress intensity factors within 1 % of ASTM E399, consistent with published fracture toughness of MgO–C. Descriptor and image-based neural networks trained on the simulations reproduce the grade ranking (Spearman 0.90–1.00) in milliseconds.

1. Introduction

The working lining of a basic oxygen furnace (BOF) is exposed to steel at 1650–1700 °C, basic FeO-rich slag and one severe thermal cycle per heat, with tap-to-tap times of about 40 min. Carbon-bonded MgO–C bricks dominate this application, although their strength is low and their carbon oxidises. Materials selection for such linings still relies on handbook properties and thermal-shock indices [1, 2]: the critical temperature difference $R = \sigma_f(1-\nu)/(E\alpha)$, its conduction-weighted form $R' = Rk$, and the damage-resistance parameter $R'''' = E\gamma/[\sigma_f^2(1-\nu)]$. Each index treats a porous multiphase brick as a homogeneous solid.

In the first part of this work we built a data-driven materials ranking ("MaterialRanger") in which machine-learning (ML) models trained on density functional theory (DFT) data supply phase properties, porosity laws [3, 4] convert them into brick properties, and the TOPSIS method [5] ranks 66 refractories against nine weighted criteria. With default weights MgO–ZrO_2 ranks first and MgO–C only fifth to seventh; MgO–C becomes the winner when the weight of R' exceeds 0.28 (Section 3). The selection therefore hinges on a quantity, the response of the microstructure to thermal shock, that the indices do not describe.

Microstructure-resolved simulations of refractories have shown that thermal-shock damage depends on phase contrast, interfaces and porosity. Özdemir et al. [6] simulated thermal shock of an idealised two-phase refractory with cohesive interfaces and continuum damage, and Damhof et al. [7] combined a calibrated damage model with thermo-elastic finite elements to predict cracking of steelmaking linings at the component scale. Two practical obstacles remain for screening many candidate microstructures: incremental–iterative solvers for quasi-brittle softening are costly and fragile near snap-back, and local strength and fracture parameters at the grain scale cannot be measured directly.

This work addresses both obstacles and links the ranking to the microstructure. The individual numerical techniques (Kanapy packing, sequentially linear analysis [9, 10, 11], cohesive zones, XFEM, contour integrals) are established; the novelty lies in what their combination makes possible and in the results it produces:

(i) *Simulation-informed decision weights.* In materials selection, criterion weights are set by expert judgement or derived from the decision matrix itself, for example by entropy weighting [39, 40], and where simulation is combined with multi-criteria methods it is continuum finite-element analysis used to verify the material already selected [40]. Here the weight that decides the ranking is tested with a microstructure-resolved model; we are not aware of a previous study that uses microstructure simulation in this way. The ranking identifies the weight of thermal-shock damage as decisive; the microstructure model shows that the industrially preferred MgO–C is the most damage-tolerant grade, which justifies a high weight on physical grounds.

(ii) *Calibration by exact scaling laws.* The strength scale and the creep law enter the model through exact homogeneity relations, so that a five-parameter calibration on open brick data reduces to one-dimensional problems with one simulation per realization, and the calibration uncertainty is propagated exactly to the predictions.

(iii) *Cross-solver verification of SLA for thermal shock.* Identical decks solved with SLA and with concrete damaged plasticity agree to within 0.016 in post-shock stiffness up to 100 K, and diverge beyond it in a way that is explained by the unloading description; this delimits the range in which SLA results for thermal shock are solver-independent.

(iv) *Microstructural separation of crack initiation and damage resistance.* Hasselman's distinction between R and R'''' is resolved in the microstructure: MgO–C has the lowest onset (26 K) but the highest retained stiffness of the MgO-based families (0.90 after 100 K), because the compliant, low-expansion graphite network arrests the cracks it nucleates.

(v) *Three-dimensional check of the two-dimensional result with all interfaces resolved.* A cohesive RVE with 13 645 interface elements and a 3-D creep RVE of the top-ranked grade confirm the 2-D conclusions and quantify the bias of the plane-stress model: 3–4 times more residual stress after severe shocks and 3.5 times less creep in 3-D, with the grade ranking unchanged.

(vi) *Emergent creep behaviour.* Load transfer from the creeping MgO to the connected graphite skeleton produces a primary-like decrease of the creep rate ($\varepsilon \propto t^{0.6}$) and an apparent activation energy of about 160 kJ/mol, well below the 281 kJ/mol of the constituent law, without any primary-creep term in the model.

(vii) *Physics-constrained image surrogates.* A periodic, translation-equivariant dual-head U-Net predicts crack and creep maps of unseen microstructures and reproduces the ranking of the five grades in milliseconds.

2. Workflow

Figure 1 summarises the chain. The ranking (blue) identifies the deciding criterion and the candidate grades. The two-dimensional microstructure model (orange) is verified, calibrated and validated, produces the dataset for the surrogates (purple), and ranks the grades. The three-dimensional models (green) test the mechanism without the plane-stress assumption and transfer the result to the brick scale. Verification, calibration and validation are used in the sense of Oberkampf and Roy [12]: verification asks whether the equations are solved correctly (exact solutions, an independent solver), calibration fits unknown parameters to measurements, and validation compares predictions with measurements that were not used in the calibration.

3. Data-driven ranking of BOF lining materials

3.1. Phase properties from machine learning

About 30 000 DFT and experimental records were collected from the Materials Project [13], OQMD [14], AFLOW [15], Alexandria [16] and curated experimental sources. Compositions are described by Magpie features [17]. A tuned gradient-boosting model [18] predicts the Voigt–Reuss–Hill shear modulus with a hold-out mean absolute error (MAE) of 0.088 dex in $\log_{10} G$ (5-fold cross-validation 0.096 dex), i.e. a typical factor of $10^{0.088} = 1.22$. This is level with the best published composition-only model in Matbench (0.099 dex [19]); structure-based graph networks reach about 0.07 dex but require a crystal structure [20] (Fig. 2a). The bulk modulus is predicted with $R^2 = 0.94$, and the thermal conductivity by transfer learning from the elastic model. On 1139 Materials Project oxides never used in training the MAE is 0.217 dex, and an unseen chemical family increases the error by about 24 %; the models therefore rank materials reliably (Spearman $\rho = 0.92$ between the ML-based and the handbook thermal-shock ranking of 17 refractory phases), whereas absolute values for design need DFT or measurements.

Error measures are used as follows: $\mathrm{MAE} = N^{-1}\sum_i |y_i - \hat{y}_i|$ in the units of the quantity; dex denotes the MAE of $\log_{10}$ values for quantities spanning decades (moduli, creep strains); $R^2 = 1 - \sum_i (y_i - \hat{y}_i)^2 / \sum_i (y_i - \bar{y})^2$ is the share of the variance explained; and Spearman's ρ compares orders.

3.2. Brick properties and TOPSIS

Phase properties are combined with porosity laws [3, 4] into brick properties and the Hasselman indices [2]. A hard filter (compatibility with basic BOF slag and service temperature $\geq$1680 °C) reduces 66 candidates to 24. TOPSIS [5] then ranks them by their relative closeness to an ideal point. Each criterion j is transformed to $\log_{10}$, the sign is inverted for criteria to be minimised, and the column is scaled to $[0, 1]$ over the candidates (x_{ij}). With weights w_j,

$$v_{ij} = w_j \frac{x_{ij}}{\sqrt{\sum_i x_{ij}^2}}, \quad d_i^{\pm} = \sqrt{\sum_j \left(v_{ij} - v_j^{\pm}\right)^2}, \quad C_i = \frac{d_i^-}{d_i^+ + d_i^-}, \tag{1}$$

where v_j^+ and v_j^- are the best and worst weighted values. Default weights are: damage resistance R'''' 0.25, BOF slag corrosion 0.25, thermal-shock resistance R' 0.15, creep 0.10, cost 0.08, oxidation, strength and manufacturability 0.05 each,

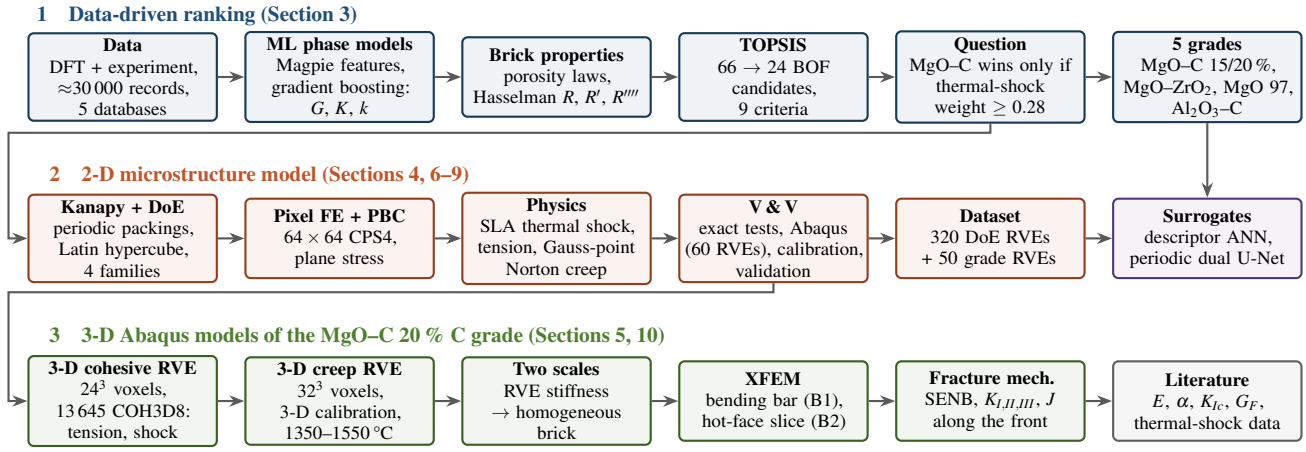
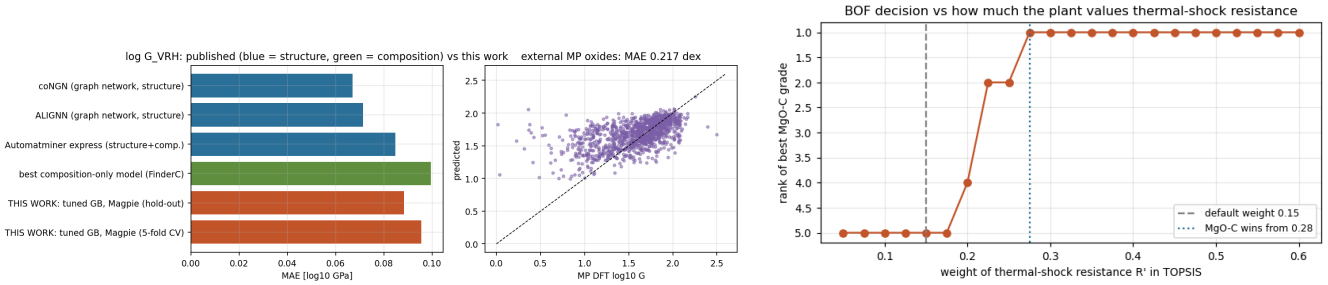


Figure 1. Workflow of the study. Blue: data-driven ranking that defines the question and the candidate grades. Orange: two-dimensional microstructure model with verification and validation (V & V). Purple: machine-learning surrogates. Green: three-dimensional Abaqus models and comparison with literature data. All two-dimensional microstructures share one pixel grid that serves as microstructure, finite-element mesh and network input.



(a) (b)

Figure 2. (a) Shear-modulus model: MAE of $\log_{10} G$ for published structure-based (blue) and composition-only (green) models and for the present composition-only model (orange), and test on 1139 Materials Project oxides not used in training. (b) Rank of the best MgO–C grade in the TOPSIS ranking as a function of the weight of thermal-shock resistance R' (other weights rescaled); dashed: default weight 0.15; MgO–C ranks first from 0.28.

Table 1. TOPSIS example for the two contrasting grades: normalised criterion scores (1 = best of the 24 candidates), distances and closeness.

Criterion	w	MgO–ZrO_2	MgO–C 20 % C
R' thermal-shock resistance	0.15	0.14	0.93
R'''' damage resistance	0.25	0.91	0.72
BOF slag resistance	0.25	1.00	0.36
Creep resistance	0.10	0.43	0.57
Oxidation resistance	0.05	0.36	0.15
Strength	0.05	0.60	0.49
Manufacturability	0.05	0.35	0.88
Cost	0.08	0.34	0.66
CO_2	0.02	0.61	0.64
d^+ / d^-		0.057 / 0.128	0.077 / 0.088
Closeness C (rank)		0.692 (1)	0.534 (7)

CO_2 footprint 0.02.

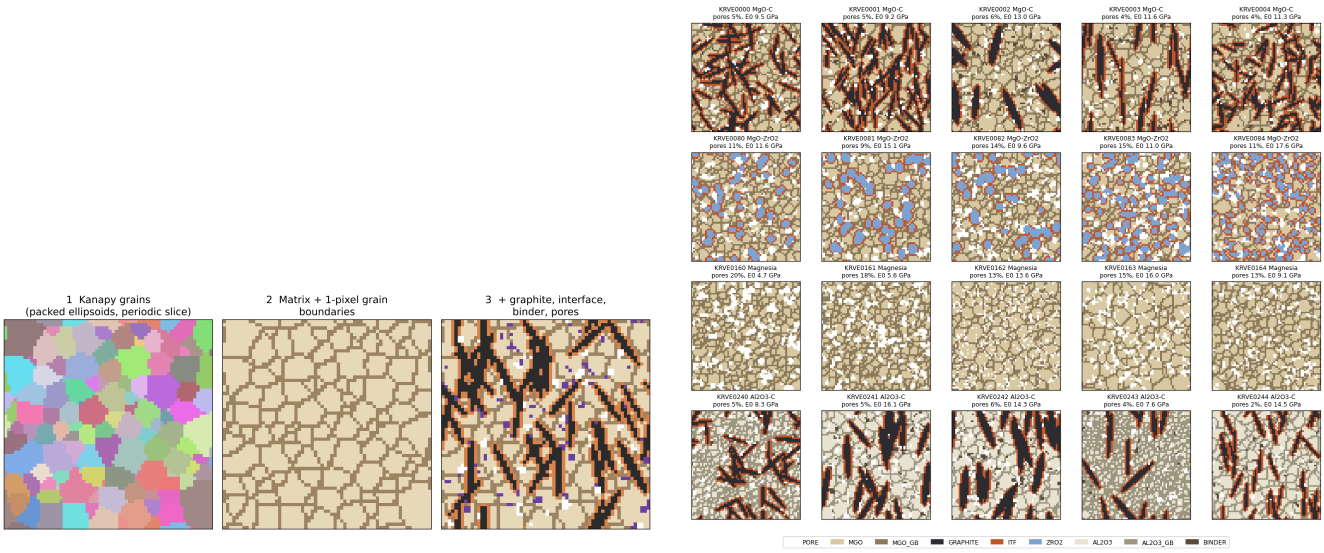
The default ranking is MgO–ZrO_2 (0.692), MgO–ZrO_2 with Y_2O_3 coating (0.673), magnesia 97 % (0.662), magnesia–chromite (0.656) and the MgO–C grades (0.534–0.592). Table 1 shows why: MgO–C has the best thermal-shock score but a low slag score, which alone accounts for 79 % of its squared distance to the ideal point, whereas the largest gap of MgO–ZrO_2 is thermal shock (59 %). Under ±30 % random perturbations of all weights (500 samples) MgO–ZrO_2 wins in 74 % and magnesia in 25 % of the cases, while MgO–C ranks first once the R' weight reaches 0.28 (Fig. 2b). Since BOF working linings are in practice made of MgO–C, the question is whether a high weight of thermal-shock damage is physically justified. The following sections answer it with a microstructure model of the five grades listed in Fig. 1.

4. Two-dimensional microstructure model

4.1. Microstructure generation

Each RVE is a periodic square of edge $L = 1$ mm discretised by $N \times N = 64 \times 64$ pixels ($h = 15.625\,\mu$m). Grain structures are generated with Kanapy [8]: equiaxed ellipsoids with log-normal equivalent diameters (shape parameter 0.3, scale d_g, cut-offs $0.5d_g$ and $2d_g$) are packed in a periodic box $L \times L \times 2.5d_g$ by Kanapy's collision-driven growth algorithm, and the mid-plane $z = L_z/2$ is voxelised periodically with the same growth criterion: a pixel centre $\mathbf{p}$ belongs to the grain with the smallest scaled distance $r = \|\mathbf{A}^{-1}\mathbf{R}^T(\mathbf{p} - \mathbf{c})\|$ over all 27 periodic images, where $\mathbf{c}$, $\mathbf{A}$ and $\mathbf{R}$ are the centre, semi-axes and rotation of the ellipsoid. Across 12 packings the grain continuity at the box faces (0.75) is close to the interior value (0.81), and the periodic slice adds only 0.20 % grain-boundary pixels.

Kanapy provides the grain structure of the matrix. The refractory phases are added in two further steps (Fig. 3a): a one-pixel grain boundary (GB) wherever the right or upper neighbour belongs to another grain, so that boundaries carry their own stiffness, strength and creep rate; then graphite flakes as periodic ellipses with mean normal along the constrained x axis (orientation $\theta \sim \mathcal{N}(90°, \sigma_\theta)$), binder and ZrO_2 particles in matrix pixels, an interface (ITF) layer on matrix pixels adjacent to graphite or ZrO_2, and pores until



(a) (b)

Figure 3. (a) Construction of an RVE: Kanapy grain packing (periodic slice), matrix with one-pixel grain boundaries, and the added graphite, interface, binder and pores. (b) Examples of the 320 design RVEs (rows: MgO–C, MgO–ZrO_2, magnesia, Al_2O_3–C; white: pores).

Table 2. Phases and calibrated properties. GB and ITF inherit ν and α of the matrix; their stiffness is $s_b E_{matrix}$ and their strength and fracture energy are reduced by k_{gb} or k_{itf}. f_t and G_F include k_f and k_f^2. Creep prefactor relative to MgO; graphite and binder are elastic.

Phase	E [GPa]	ν	α [10^{-6}/K]	f_t [MPa]	G_F [N/m]	creep A/A_{MgO}
MgO	300	0.18	13.5	64.0	12.8	1
MgO GB	3.11	0.18	13.5	$k_{gb}\times$	$k_{gb}\times$	3
ITF	3.11	0.18	13.5	$k_{itf}\times$	$k_{itf}\times$	1
Graphite	15	0.20	3.0	20.0	32.0	–
ZrO_2	210	0.30	10.0	96.1	51.3	5
Al_2O_3	380	0.23	8.0	96.1	16.0	0.2
Al_2O_3 GB	3.93	0.23	8.0	$k_{gb}\times$	$k_{gb}\times$	0.6
Binder	20	0.20	4.0	12.0	6.4	–

the target porosity is reached. Solid islands not connected to the periodic skeleton are removed by union–find, since they would make the stiffness matrix singular. Phase properties are listed in Table 2.

4.2. Design of experiments

Three sets of microstructures are used. (a) 320 design RVEs, 80 per family, chosen by a Latin-hypercube design [21]: each factor range is divided into 80 equal intervals, each interval is used exactly once, and the intervals are paired randomly between factors (Table 3). With 6–11 factors per family, a full-factorial design with only three levels would need up to $3^{11} = 177\,147$ runs per family. Each design RVE also receives a creep stress and temperature from the design; thermal shock is evaluated at fixed levels of 25, 50, 100, 200 and 400 K. (b) 10 realizations of each of three literature bricks for calibration and validation. (c) 10 realizations of each of the five ranked grades (MgO–C with 15 and 20 % C, MgO–ZrO_2, magnesia 97, Al_2O_3–C); only the random seed differs between realizations.

4.3. Finite-element model and periodic boundary conditions

Each solid pixel is a four-node bilinear plane-stress element with 2×2 Gauss integration (Abaqus CPS4); pores are not meshed, giving 3300–3900 elements and 7600–8200 unknowns per RVE. Because all elements are identical squares,

Table 3. Latin-hypercube design: factors and ranges (uniform sampling), 80 runs per family, seed 2026.

Factor	MgO–C	MgO–ZrO_2	Magnesia	Al_2O_3–C
Second phase fraction	0.10–0.30	0.05–0.25	–	0.10–0.25
Flake length [µm]	120–400	–	–	120–400
Flake aspect ratio	5–12	–	–	5–12
Orientation scatter [rad]	0.10–1.20	–	–	0.10–1.20
Particle / pore size [µm]	–	30–90	20–60	–
Porosity	0.02–0.08	0.08–0.18	0.10–0.20	0.02–0.08
Grain size [µm]	60–160	60–160	60–160	40–120
Binder fraction	0–0.05	–	–	0–0.05
k_{gb}	0.40–0.80	0.40–0.80	0.40–0.80	0.40–0.80
k_{itf}	0.15–0.60	0.50–1.00	–	0.15–0.60
Creep stress [MPa]	1–8	1–8	1–8	1–8
Creep temperature [°C]	1350–1550	1350–1550	1350–1550	1350–1550
Factors / runs	11 / 80	8 / 80	6 / 80	11 / 80

the element stiffness is $\mathbf{K}_e = \lambda_e^* \mathbf{K}_\lambda + \mu_e \mathbf{K}_\mu$ with two precomputed matrices and $\lambda^* = E\nu/(1-\nu^2)$, $\mu = E/[2(1+\nu)]$. Periodicity is imposed by

$$\mathbf{u}(L,y) = \mathbf{u}(0,y) + \mathbf{U}^{(1)}, \qquad \mathbf{u}(x,L) = \mathbf{u}(x,0) + \mathbf{U}^{(2)}, \quad (2)$$

with the macroscopic strains $\bar{\varepsilon}_{xx} = U_x^{(1)}/L$, $\bar{\varepsilon}_{yy} = U_y^{(2)}/L$ and $2\bar{\varepsilon}_{xy} = (U_y^{(1)} + U_x^{(2)})/L$. Dependent degrees of freedom are eliminated exactly through $\mathbf{u} = \mathbf{T}\mathbf{q}$, giving $\mathbf{T}^T\mathbf{K}\mathbf{T}\mathbf{q} = \mathbf{T}^T\mathbf{f}$; the same equations are written to Abaqus as `*EQUATION`. Macroscopic stresses follow from the reactions of the two reference points that carry $\mathbf{U}^{(1)}, \mathbf{U}^{(2)}$, divided by the edge area [22]. One corner node is fixed and $U_y^{(1)} = 0$ removes the rigid rotation.

Load cases (Fig. 4). Thermal shock is modelled as constrained cooling of the surface layer [1]: an eigenstrain $\varepsilon^* = -\alpha\Delta T(1,1,0)$ with $\bar{\varepsilon}_{xx} = 0$ along the face and $\bar{\sigma}_{yy} = 0$ into the depth. For a homogeneous solid this gives $\sigma_{xx} = E\alpha\Delta T$ (4.050 MPa/K for MgO). Creep is a constant macroscopic stress $\bar{\sigma}_{xx} = -\sigma$ with $\bar{\sigma}_{yy} = 0$ held for 10 h; uniaxial tension uses $\bar{\varepsilon}_{xx}$ as load factor with $\bar{\sigma}_{yy} = 0$.

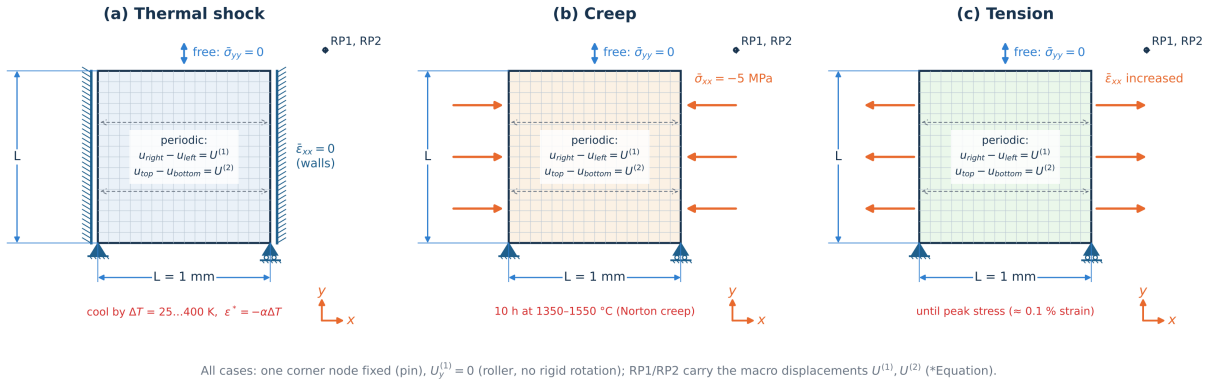


Figure 4. Two-dimensional RVE model: geometry, periodic boundary conditions and the three load cases. (a) Thermal shock: x (in the hot face) held, y (into the brick) free, cooling as eigenstrain. (b) Creep: $\bar{\sigma}_{xx} = -5$ MPa for 10 h at 1350–1550 °C (grades). (c) Uniaxial tension to the peak stress. RP1/RP2 carry the macroscopic displacements $\mathbf{U}^{(1)}, \mathbf{U}^{(2)}$.

4.4. *Sequentially linear analysis with an energy-consistent saw-tooth law*

The SLA replaces the softening problem by a sequence of linear problems with a damaged secant stiffness [9, 10]. For the current moduli $E_e = (1-d_e)E_{0,e}$ the stresses for $\Delta T = 1$ K are computed; since stresses are proportional to ΔT between events, the critical temperature drop of each element is

$$\Delta T_e = f_{e,k} / \sigma_{1,e}^{(1\,\mathrm{K})}, \tag{3}$$

where σ_1 is the maximum principal stress and $f_{e,k}$ the strength on tooth k. The element with the smallest ΔT_e moves to the next tooth and the system is solved again. Under load control all events below a requested level are processed, so unstable crack growth appears as a cascade at constant ΔT. Events within 5 % of the critical load (30 % in deep cascades) are processed together; 5–6 times finer batches changed $\log_{10} E/E_0$ by a median of 0.002, against an inter-microstructure scatter of 0.071.

Each element follows a linear tension-softening law with ultimate strain $\varepsilon_u = 2G_F/(f_t h)$, i.e. a crack band of width h [23, 24], which makes the dissipated energy independent of the element size. Tooth k has secant modulus $E_k = (1-d_k)E_0$ and strength

$$f_k = \frac{E_k f_t \varepsilon_u}{E_k(\varepsilon_u - \varepsilon_0) + f_t}, \qquad \varepsilon_0 = f_t/E_0. \tag{4}$$

The energy dissipated per crack area along the saw-tooth path is

$$W = h \sum_k \tfrac{1}{2} \varepsilon_k \left(f_k - E_{k+1}\varepsilon_k \right), \qquad \varepsilon_k = f_k/E_k. \tag{5}$$

For brittle phases the softening branch is long ($\varepsilon_u/\varepsilon_0 \approx 120$ for MgO), and a discretisation with few damage levels dissipates only a small fraction of G_F. We therefore use 25 geometric teeth, $E_k = 0.7^k E_0$, and rescale ε_u per element by the fixed-point iteration $\varepsilon_u \leftarrow \varepsilon_u G_F/W(\varepsilon_u)$ until $W = G_F$ (Fig. 5), following the regularisation of Rots and Invernizzi [11]. Recorded at each level are the damage map, the retained uniaxial modulus E/E_0, the secant ratio $\bar{\sigma}_{xx}/(E_0 \alpha_{eff} \Delta T)$ and the onset ΔT_c (first event).

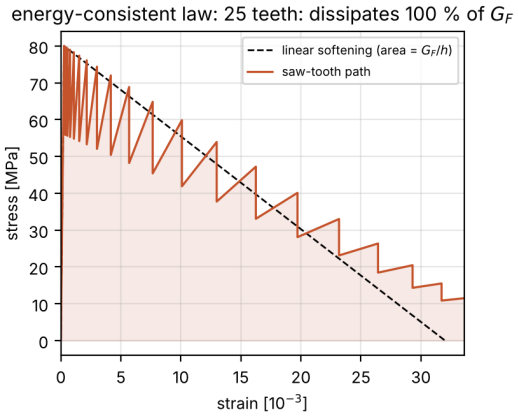


Figure 5. Energy-consistent saw-tooth law for MgO: 25 teeth with $E_k = 0.7^k E_0$ and per-element rescaled ultimate strain, so that the shaded area (energy per volume) equals G_F/h of the linear softening law (dashed).

4.5. *Creep*

Each creeping phase follows the von Mises–Norton law

$$\dot{\varepsilon}^{cr} = \tfrac{3}{2} A(T)\, q^{n-1} \mathbf{s}, \quad A(T) = A_{ref}\, r_{ph}\, g\, e^{-\frac{Q}{R}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)}, \tag{6}$$

with von Mises stress q, deviator $\mathbf{s}$, phase ratio r_{ph} relative to MgO, grain-boundary enhancement $g = 3$ and $T_{ref} = 1550$ °C. Creep strains are stored at the four Gauss points, so that the stress peaks next to pores creep at their own rate, and integrated explicitly with the elastic stiffness factorised once; the time step is limited so that the creep increment does not exceed 20 % of the elastic strain. If all creeping phases share the exponent n, the problem is homogeneous and

$$\varepsilon^{cr}(t;\sigma,A) = \sigma\, e_1\left(\sigma^{n-1} \tfrac{A}{A_b} t\right), \tag{7}$$

where e_1 is the response to 1 MPa with prefactor A_b. One master curve per microstructure therefore gives the creep strain for any stress, prefactor and temperature.

4.6. *Machine-learning surrogates*

A descriptor network (23 inputs: family, phase fractions, flake and grain parameters, k_{gb}, k_{itf}, homogenised E_0 and α_{eff}, creep stress and temperature with $\log\sigma$ and $1/T$, and

the homogenised Hasselman index; two hidden layers of 64 ReLU units) predicts nine logarithmic targets (onset, E/E_0 at five levels, tensile strength, creep rate and strain). It is evaluated by 5-fold and leave-one-family-out cross-validation against a random forest and a Hasselman-index regression. A dual-head U-Net [25] with periodic padding shares one encoder (16–128 filters) between a crack decoder conditioned on ΔT and a creep decoder conditioned on (σ, T, t); inputs are one-hot phase channels of the 64×64 pixel map. Data are split by whole RVE (70/15/15 %), and periodic shifts and mirrors are used for augmentation; rotations are excluded because they exchange the constrained and free directions.

5. Three-dimensional Abaqus models

The three-dimensional models serve as an independent check of the two-dimensional result for the grade that the 2-D model identified as the most damage-tolerant MgO–C grade, MgO–C 20 % C. They remove the two main simplifications of the 2-D model, plane stress and a hot face held in only one direction, and they use a different crack description (cohesive interfaces and XFEM instead of smeared saw-tooth damage) and a different solver. The models are solved with Abaqus/Standard 2025 (units mm, N, MPa, s, K). Inputs are flat input decks generated in Python; each deck is checked by a static validator and by an independent linear re-solution before submission.

5.1. Cohesive RVE (A1 tension, A2 thermal shock)

A Kanapy cube of edge 0.5 mm (seed 20) is voxelised with 24^3 voxels ($h = 20.8\,\mu$m): graphite fraction 0.28, flake length 250 µm, aspect ratio 8, orientation scatter 0.40 rad, porosity 0.04, grain size 130 µm, binder 0.03, $k_{gb} = 0.6$, $k_{itf} = 0.3$ (Fig. 6). Each grain piece, graphite network and binder blob (607 bodies) receives its own nodes, and every interface between bodies is filled with zero-thickness cohesive elements (COH3D8): 4928 grain-boundary, 7112 graphite-interface and 1605 binder-interface elements, together with 13 271 C3D8 bulk elements and 34 935 nodes. Bulk phases are elastic; cracking is confined to the interfaces. The calibrated 2-D boundary layer is mapped onto the interfaces: normal stiffness $K_n = s_b E_{MgO}/h_{2D} = 1.99 \times 10^5$ N/mm^3 and $K_s = K_n/[2(1+\nu)]$. The traction–separation law is bilinear,

$$t = (1-D)\,K\delta, \qquad \left(\frac{\langle t_n\rangle}{t_n^0}\right)^2 + \left(\frac{t_s}{t_s^0}\right)^2 + \left(\frac{t_t}{t_t^0}\right)^2 = 1, \quad (8)$$

with quadratic-traction initiation and Benzeggagh–Kenane energy propagation [26] ($\eta = 2$, $G_{II} = G_{III} = 2G_I$). Strengths and energies follow the calibrated 2-D values: t^0 = 38.4, 19.2 and 12.0 MPa and G_{Ic} = 7.7, 3.8 and 6.4 J/m^2 for grain boundaries, graphite interfaces and binder interfaces. Element deletion is off, the maximum degradation is 0.99 and a viscous regularisation of 2×10^{-4} is used.

Boundary conditions (Fig. 7): the faces $x = 0$, $y = 0$, $z = 0$ are symmetry planes. In A1 the face $x = L$ is tied to a reference point that moves to $\bar{\varepsilon}_{xx} = 0.4\,\%$. In A2 the faces $x = L$ and $z = L$ are held by reference points, y is free, and the cube is cooled uniformly by up to 400 K: a hot-face element held in both in-face directions, which the 2-D model cannot represent.

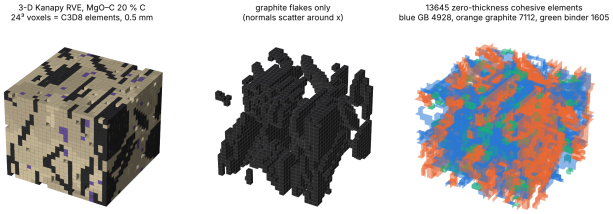


Figure 6. 3-D Kanapy RVE of MgO–C 20 % C (24^3 voxels, 0.5 mm): microstructure, graphite networks, and the 13 645 cohesive elements (blue grain boundaries, orange graphite interfaces, green binder interfaces).

5.2. Creep RVE (C1–C3)

A 32^3 voxel model of the same grade (0.5 mm, 31 457 C3D8) contains a grain-boundary layer defined on the 26-voxel neighbourhood, which closes the boundary network in three dimensions. Its boundary stiffness and creep law were recalibrated in 3-D on the same data as in 2-D: $s_b = 0.0207$ from $E = 12$ GPa of the MgO–C 15 % brick [27], and $A_{ref} = 4.33 \times 10^{-8}$ MPa^{-n}s^{-1} and $Q = 281$ kJ/mol at $n = 1.75$ from the magnesia creep tests at 1350 and 1425 °C [28]. A compressive load of 5 MPa is applied in a static step and held for 10 h (`*VISCO`) at 1350, 1450 and 1550 °C. Graphite and binder do not creep.

5.3. Brick-scale models (B1, B2, D1) and the two scales

A fully resolved brick is out of reach: a $150 \times 25 \times 25$ mm bar at 21 µm voxels would need about 10^{10} elements, whereas the 0.5 mm RVE already requires 5 h on 8 cores. The study therefore uses two scales (Table 4). The RVEs resolve where and why cracks start in the microstructure; the brick-scale models use a homogeneous material with the RVE stiffness ($E = 13.4$ GPa, $\nu = 0.15$, $\alpha = 1.11 \times 10^{-5}$ K^{-1}) and brick-level fracture parameters (tensile strength $f_t = 11.5$ MPa, below the measured modulus of rupture of 13 MPa [27]; fracture energy $G_F = 150$ J/m^2, between the value implied by the measured fracture toughness of MgO–20 % C, $G = K_{Ic}^2(1-\nu^2)/E = 116$ J/m^2 [37], and the wedge-splitting fracture energy of MgO–C at room temperature, 240 J/m^2 [29]). The RVE strength cannot be transferred directly because a 0.5 mm cube contains no millimetre flaws.

- **B1, three-point bending with XFEM** (Fig. 8a): bar $150 \times 25 \times 25$ mm, span 125 mm, $2.5 \times 1.56 \times 2.5$ mm C3D8 elements (9600), displacement-controlled load line ($u_y = -0.3$ mm). An enriched zone of ±20 mm around mid-span allows a crack to start anywhere when the maximum principal stress reaches f_t, with linear softening governed by G_F and damage stabilisation 5×10^{-5} [30].
- **B2, hot-face slice with XFEM** (Fig. 9): $50 \times 30 \times 10$ mm slice, sides held ($u_x = 0$), cold back face held ($u_y = 0$), symmetry at $z = 0$; 2 mm elements in x and z, 0.83 mm in y within 10 mm of the hot face (2750 C3D8). The slice starts stress-free at operating temperature; the hot face $y = 0$ is cooled by $\Delta T_s = 300$ K with the profile $\Delta T(y) = -\Delta T_s\,\mathrm{erfc}\left(y/2\sqrt{\kappa t}\right)$, $\kappa = 3$ mm^2/s, $t = 10$ s (penetration about 11 mm). Three enrichment strips (at $x = 11$, 25 and 39 mm, strength ±4 % in five groups) let cracks compete.
- **D1, single-edge notched beam** (Fig. 8b): ASTM E399 geometry [31], $W = B = 25$ mm, $S = 4W$, seam crack

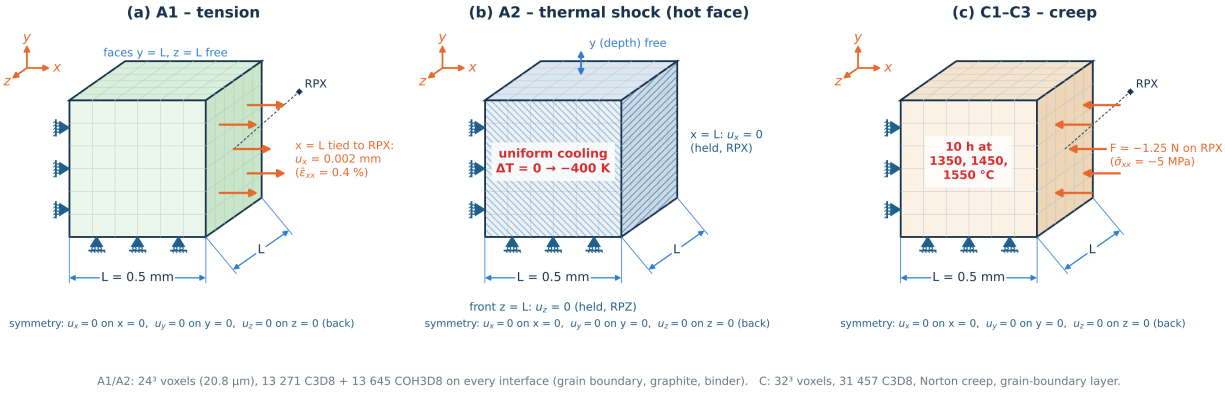
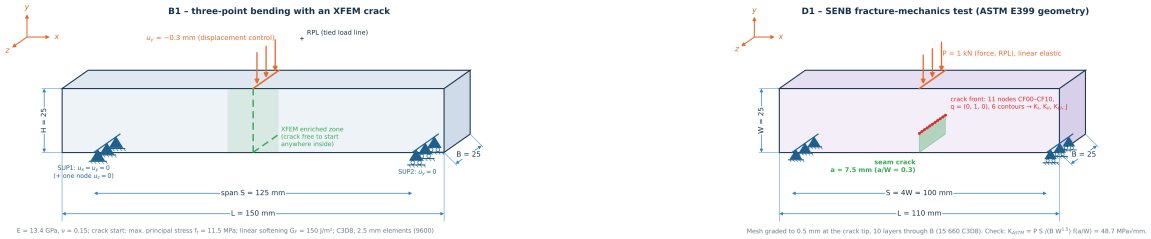


Figure 7. Three-dimensional RVE models: symmetry planes on $x = 0$, $y = 0$, $z = 0$ and reference points (RPX, RPZ) on the opposite faces. (a) A1 tension, $u_x = 0.002$ mm ($\bar{\varepsilon}_{xx} = 0.4\,\%$). (b) A2 thermal shock, x and z held, depth y free, uniform cooling to -400 K. (c) C1–C3 creep, $F = -1.25$ N ($\bar{\sigma}_{xx} = -5$ MPa) for 10 h.



(a) (b)

Figure 8. Brick-scale models (dimensions in mm). (a) B1: three-point bending of a bar with an XFEM-enriched zone at mid-span. (b) D1: single-edge notched beam with a seam crack of 7.5 mm; the 11 crack-front nodes CF00–CF10 carry the contour integrals.

Table 4. The two scales of the study.

	Microscale: RVE (A1, A2, C1–C3)	Brick scale (B1, B2, D1)
Size	0.5 mm, 21 µm voxels	25–150 mm, 0.5–2.5 mm elements
Material	resolved grains, graphite, binder, 13 645 interfaces	homogeneous: E from the RVE; brick f_t, G_F
Answers	where and why cracks start; load sharing between phases	where and at which load a part cracks; crack-tip loading

$a/W = 0.3$, mesh graded to 0.5 mm at the front, 10 layers through the thickness (15 660 C3D8), $P = 1$ kN. Contour integrals [32, 33] give K_I, K_{II}, K_{III} and J at 11 front nodes over six contours.

6. Verification

6.1. Exact solutions

The two-dimensional solver reproduces closed-form results to round-off (Table 5). The scaling laws of Section 7 (strength proportional to k_f; Eq. 7) are exact and are verified directly.

6.2. Independent solution with Abaqus (2-D)

The meshes, periodic equations and material tables of 60 calibrated models (20 design RVEs and two realizations per ranked grade; 30 thermal-shock models cooled to 400 K and 30 creep models of 10 h) were written as Abaqus/Standard decks with identical node and element labels and solved independently. Thermal shock uses concrete damaged plasticity [34, 35] with linear softening defined by G_F and tension

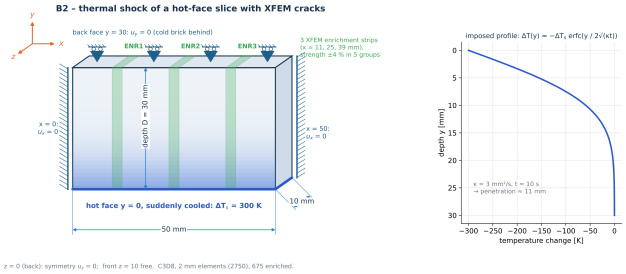


Figure 9. B2: hot-face slice with three XFEM enrichment strips; the hot face $y = 0$ is cooled by 300 K with an erfc profile (right).

Table 5. Tests with known answers (relative errors).

Test	Result
Constrained cooling $\sigma_{xx} = E\alpha\Delta T$	1.4×10^{-13}
Laminate, series and parallel rules	exact
Periodic shift invariance	0
SLA onset, homogeneous block $f_t/(E\alpha)$	19.753 K (exact)
Norton rate, homogeneous block $A\sigma^n$	5.1×10^{-13}
SLA tension: strength $= f_t$, modulus $= E$	exact (80.000 MPa)
Strength $\propto k_f$	ratio 0.370000 for $k_f = 0.37$
Creep master curve, Eq. (7)	4.2×10^{-11}
Kanapy rotation convention	1.8×10^{-15}
Periodic Kanapy slice, extra GB pixels	0.20 %

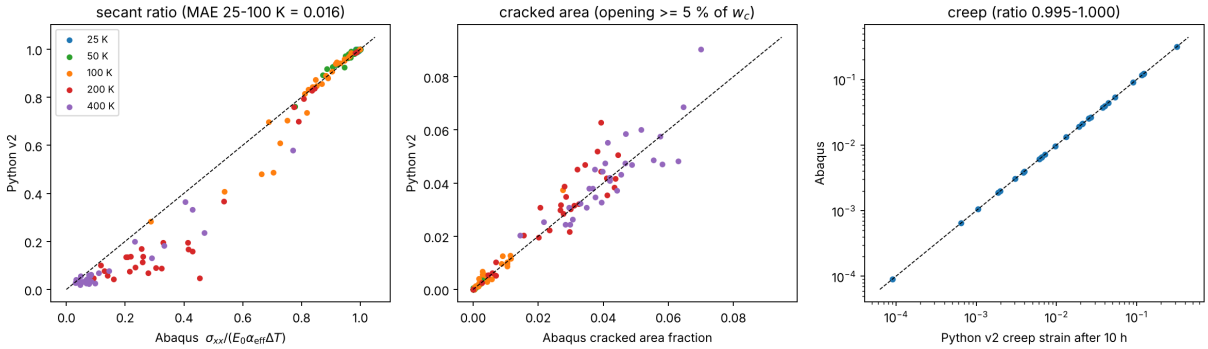


Figure 10. Calibrated 2-D model, Abaqus versus Python for 60 RVEs. Left: secant ratio at all levels. Middle: cracked area fraction. Right: macroscopic creep strain after 10 h.

Table 6. Calibrated model, Abaqus versus Python (medians over 30 thermal-shock and 30 creep models).

ΔT [K]	25	50	100	200	400
Secant, Abaqus	0.999	0.974	0.876	0.291	0.076
Secant, Python	1.000	0.982	0.876	0.136	0.040
Mean abs. difference	0.003	0.010	0.034	0.115	0.052
Cracked area [%] A/P	0/0	0/0	0.27/0.34	2.8/3.1	4.1/4.3
1-px overlap	–	0.40	0.74	0.78	0.81
Moduli: max. rel. difference 5.3×10^{-8}					
Creep strain 10 h: ratio 0.995–1.000; maps $R^2\geq0.9997$					

damage $d_t = 0.99\,w/w_c$, $w_c = 2G_F/f_t$; creep uses `*CREEP, LAW=TIME` under `*VISCO`. The decks are written in micrometre units, which keeps the plastic crack-opening tables of the very compliant boundary phases well conditioned. All analysis steps converged. A homogeneous test model reproduces the slope $E\alpha = 4.050$ MPa/K and the strength (64.30 MPa against $f_t = 64.05$ MPa; the difference is the viscous overstress of the regularisation).

With a common crack definition (opening $w \geq 0.05\,w_c$ in both codes, which excludes small elastic openings) the codes agree closely (Table 6, Figs. 10 and 11): moduli are identical, creep strains agree within 0.5 %, the secant ratio agrees within 0.016 over 25–100 K (rank correlation 0.99 at 100 K), and 74–81 % of the crack pixels of one code lie within one pixel of a crack of the other. Beyond 100 K the cracked RVE carries about twice as much residual stress in Abaqus (secant 0.29 against 0.14 at 200 K) because concrete damaged plasticity retains a plastic crack opening and a viscous stress, whereas SLA unloads secantly; the cracked areas differ by only about 10 %. Results above 100 K are therefore reported as indicative, and the grade ranking uses 100 K.

6.3. *Verification of the 3-D models*

Each 3-D model was checked against an independent quantity (Table 7). Elastic responses were computed with an in-house 3-D finite-element code on the same meshes (algebraic-multigrid preconditioned conjugate gradients); the cohesive law was checked element by element against the input; the stress intensity factor against ASTM E399 [31] and an independent compliance calculation; and the XFEM crack onset in B2 against Hasselman's critical temperature difference for a face held in two directions, $\Delta T_c = f_t(1-\nu)/(E\alpha) = 66$ K.

Microstructure scatter. The cohesive RVE was rebuilt for four further Kanapy seeds (21–24) with the same recipe, and the elastic response and the onset of interface damage were

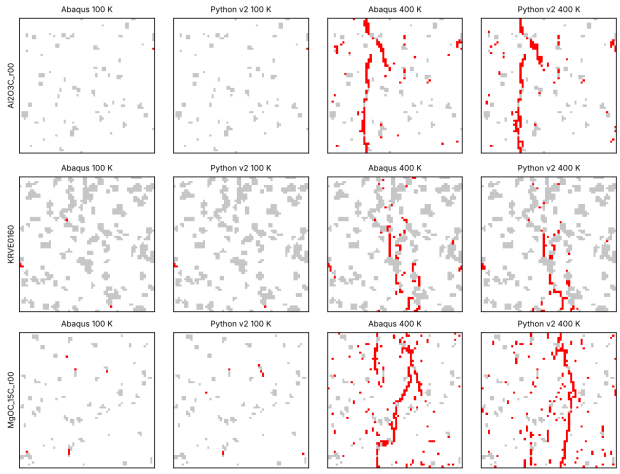


Figure 11. Crack maps (red; grey: pores) after 100 and 400 K for an Al_2O_3–C grade (top), a magnesia RVE (middle) and an MgO–C 15 % grade (bottom), Abaqus and Python side by side.

Table 7. Verification of the 3-D Abaqus models.

Quantity	Independent / input	Abaqus
E_0, cohesive RVE (A1)	13.3 GPa	13.3 GPa
Elastic hot-face stress (A2)	0.159 / 0.241 MPa/K	within 3 %
Interface peak tractions	38.4 / 19.2 / 12.0 MPa	35.9 / 18.5 / 12.0 MPa
Opening: node pair vs element	–	identical
Energy per broken area (A1)	3.8–7.7 J/m^2 (inputs)	≈6 J/m^2
XFEM crack onset (B2)	66 K (Hasselman)	72 K
K_I, mid-thickness (D1)	48.7 (ASTM), 48.9 (compl.) MPa$\sqrt{\mathrm{mm}}$	48.3 MPa$\sqrt{\mathrm{mm}}$
K_I contour spread (D1)	–	0.4 %

computed for all five realizations with the independent solver (Table 8). The coefficients of variation are 0.4–4.9 % for the stiffness, the expansion coefficient and the hot-face stresses, 7 % for the cooling at first damage and 12–15 % for the first damage in tension. In all five realizations, cooling damages a binder interface first; in tension, binder (four realizations) or graphite interfaces (one) start first.

The grain-boundary peak traction is 7 % below the input because that interface opens in mixed mode (quadratic-traction criterion); the graphite and binder interfaces reach their input strength. The crack onset in B2 lies 9 % above the hand estimate because the stresses are evaluated at element centres 0.4 mm below the surface, where the temperature drop is

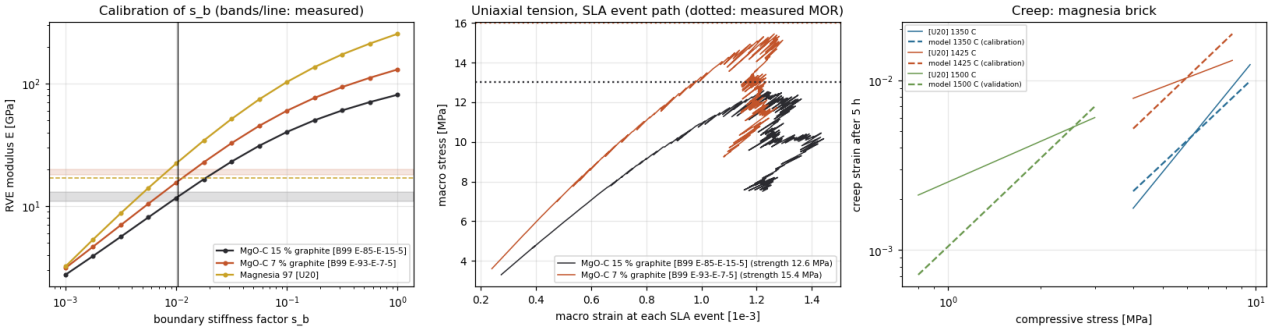


Figure 12. Calibration. Left: RVE modulus versus boundary stiffness factor for the three bricks (bands: measured). Middle: SLA stress–strain event paths in uniaxial tension (dotted: measured MOR). Right: creep strain after 5 h versus stress, measured (solid) and model (dashed); 1500 °C is a prediction.

Table 8. 3-D cohesive RVE of MgO–C 20 % C, five Kanapy realizations (independent linear solver; seed 20 is the Abaqus model).

Seed	E_0 [GPa]	α_{eff} [10^{-6}/K]	$\sigma_{xx}/\Delta T$ [MPa/K]	$\sigma_{zz}/\Delta T$ [MPa/K]	ΔT, 1st damage [K]	ε, 1st damage [%]
20	13.35	11.09	0.159	0.241	37.0	0.060
21	12.34	11.04	0.148	0.249	40.2	0.080
22	13.91	11.06	0.169	0.254	40.2	0.060
23	13.46	10.97	0.159	0.239	45.2	0.071
24	13.07	11.07	0.155	0.239	39.7	0.057
Mean	13.23	11.05	0.158	0.244	40.5	0.066
CV [%]	4.4	0.4	4.9	2.8	7.3	14.8

smaller, and the strengths of the enriched elements vary by ±4 %. The full 3-D jobs required between 2 min (B2) and 5 h 14 min (A2) on 8 cores.

7. Calibration

Elastic constants and expansion coefficients of the dense phases are handbook values, consistent with the DFT moduli within about 5 %; grain-scale strengths, fracture energies and creep constants are priors. Five parameters are calibrated (Fig. 12): (1) the boundary stiffness factor s_b ($E_{GB} = E_{ITF} = s_b E_{matrix}$), which represents microcracks, fine porosity and weak bonds below the pixel size, fitted by bisection to $E = 12$ GPa of an MgO–C brick with 15 wt% graphite [27]; (2) a strength scale k_f applied to all f_t, with $G_F \propto k_f^2$ so that Hillerborg's characteristic length EG_F/f_t^2 is unchanged; then every strain and strength of the saw-tooth law scales with k_f and the SLA strength scales exactly with k_f, so k_f follows from one simulation per realization and the modulus of rupture (13 MPa) of the same brick; and (3–5) A_{ref}, n and Q, obtained by least squares in log strain against the creep strains after 5 h of a fired magnesia brick at 1350 and 1425 °C [28], computed from the reported Norton–Bailey law $\varepsilon = [(1-a)K\sigma^n t]^{1/(1-a)}$, using Eq. (7) for $n \in [1,3]$. Graphite weight fractions were converted to volume fractions with theoretical densities (15 wt% → 22.2 vol%).

Before calibration the RVE moduli of the three bricks were 82–255 GPa, 7–15 times the measured values, because dense-phase properties ignore the compliance of microcracks and weak bonds. The calibration gives $s_b = 0.0104$ ($E_b = 3.1$ GPa), $k_f = 0.801$ (MgO $f_t = 64$ MPa), $n = 1.75$, $Q = 274$ kJ/mol and $A_{ref} = 4.87 \times 10^{-8}$ MPa^{-n}s^{-1}, with an rms error of 0.053 dex for the six creep points. The calibration uncertainty, propagated exactly through the scaling laws, is ±7.8 % in k_f and ±0.05 dex in the creep prefactor.

8. Validation and comparison with literature

8.1. Held-out brick data

Held out for validation, with acceptance bands fixed beforehand (±25 % for stiffness and strength ratio, ±0.3 dex for creep), were the modulus and strength ratio of a brick with 7 wt% graphite [27], the modulus of the magnesia brick [28] and its creep at 1500 °C (Table 9, Fig. 13). The effect of graphite is predicted without refitting: the modulus of the 7 % brick within −16 % and the strength ratio of the two bricks within −6 %. Creep at 1500 °C is predicted within 0.01 and 0.22 dex at 2.5 and 1.5 MPa.

Magnesia modulus. The porosity of the magnesia brick is not reported in [28]; the value of 5 % used in the calibration set follows from the density of 3400 kg/m^3 that [28] lists as a finite-element input, whereas fired refractories typically contain up to 20 vol% porosity [28]. With 5 % porosity the prediction is +30 % above the measured 17 GPa. Because the porosity is an unmeasured input rather than a model parameter, its effect was quantified without changing the calibration (Fig. 14): the predicted modulus falls linearly from 22.5 to 11.2 GPa between 5 and 20 % porosity, lies within the acceptance band for porosities of 6.5–17.7 % and equals the measurement at 12 %. The magnesia modulus is therefore not a discriminating validation quantity until the porosity of that brick is known, and it is reported separately in Table 9.

Creep at 1 MPa. The 5 h creep strains of [28] imply apparent stress exponents of 2.2 at 1350 °C but 0.7–0.8 at 1425 and 1500 °C, i.e. a stress sensitivity that changes with temperature and stress, consistent with a transition towards diffusion-controlled creep ($n \approx 1$) at low stresses. A single Norton law with $n = 1.75$, calibrated at 5–8 MPa, cannot follow this change; it is accurate at 2.5 MPa (−0.01 dex) and underestimates creep at 1.0 MPa (−0.38 dex), a stress five times below the calibration range. The creep model is therefore stated as valid for $\sigma \geq 2.5$ MPa in the calibrated temperature window, which includes the 5 MPa used for the ranked grades. Identifying a second, linear mechanism requires creep data below 2.5 MPa that are independent of the validation set.

8.2. Comparison with published fracture and thermal-shock data

Measured retained properties after quenching of the investigated grades were not available for this study. The thermal-shock and fracture predictions are therefore compared with published data of MgO–C bricks (Table 10).

The stiffness and thermal expansion of the model lie within

Table 9. Validation on data not used in calibration (mean of 10 realizations; acceptance bands fixed beforehand: ±25 %, ±0.3 dex).

Quantity	Measured	Model	Error
E, MgO–C 7 % graphite [27]	19 ± 1 GPa	16.0 GPa	−16 %
Strength ratio 7 %/15 % [27]	1.23 ± 0.12	1.16	−6 %
Creep 5 h, 1500 °C, 2.5 MPa	5.2×10^{-3}	5.1×10^{-3}	−0.01 dex
Creep 5 h, 1500 °C, 1.5 MPa	3.5×10^{-3}	2.1×10^{-3}	−0.22 dex
Outside the stated validity / input range			
E, magnesia 97, porosity 5 %[a]	17 GPa	22.1 GPa	+30 %
E, magnesia 97, porosity 12 %[a]	17 GPa	17.0 GPa	0 %
Creep 5 h, 1500 °C, 1.0 MPa[b]	2.5×10^{-3}	1.1×10^{-3}	−0.38 dex

[a] porosity not reported in [28]; see Fig. 14. [b] five times below the lowest calibration stress; apparent stress exponent of the data 0.8.

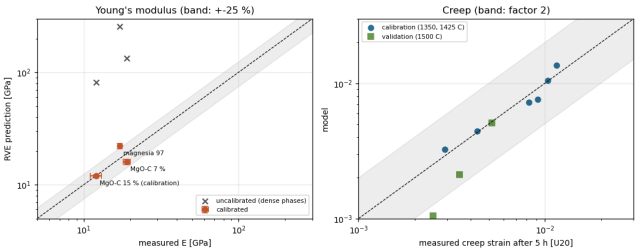


Figure 13. Validation. Left: Young's modulus, before calibration (crosses) and after calibration (circles), band ±25 %. Right: creep strain after 5 h, calibration points (1350, 1425 °C) and predictions (1500 °C), band: factor 2.

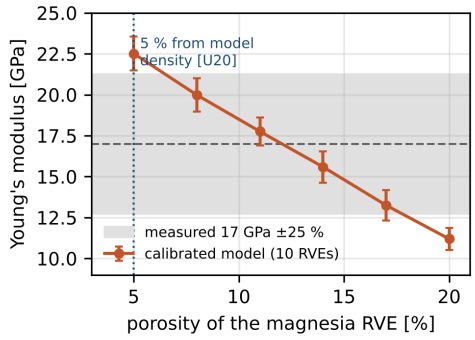


Figure 14. Calibrated model, modulus of the magnesia brick as a function of the unreported porosity (mean ± standard deviation of 10 Kanapy realizations; no parameter changed). Grey band: measured 17 GPa ±25 % [28].

the ranges measured on commercial MgO–C bricks. The fracture toughness implied by the brick parameters ($K_{Ic} = \sqrt{EG_F/(1-\nu^2)} = 1.43\,\mathrm{MPa}\sqrt{\mathrm{m}}$) is 14 % above the mean measured by Hino et al. [37] on MgO–20 % C at room temperature and within 5 % of their largest values. The critical temperature difference published for commercial bricks (171–330 K) was computed with the compressive strength [36]; with a tensile strength of 11.5 MPa the same index gives 66 K, in agreement with the XFEM onset of 72 K. A single water quench of MgO–10 % C from 1000 °C leaves 47–87 % of the bending strength [38], i.e. the material degrades gradually rather than failing abruptly, as predicted. A quantitative comparison requires the quench geometry: in a quenched bar the surface stress is reduced by the Biot-number dependent stress-reduction factor and acts only in a thin layer, whereas the RVE represents the fully constrained hot face itself.

9. Results of the two-dimensional model

9.1. Thermal shock and creep across the four families

Under constrained cooling, cracks nucleate at pores, weak interfaces and grain boundaries, run along y (normal to the constrained axis, into the depth) and coalesce into a through-crack at 200–400 K (Fig. 15). Table 11 summarises the 320 design RVEs. MgO–C cracks first (median onset 26 K) but retains the largest stiffness of the MgO-based families after 100 and 200 K. Al_2O_3–C combines a late onset with high retained stiffness owing to its lower expansion mismatch. The simulated onset lies one to two orders of magnitude below the homogenised index $f_t/(E_{eff}\alpha_{eff})$, and a regression on this index explains only $R^2 = 0.44$ of the onset and 0.26 of $\log E/E_0$ at 100 K: local stress concentrations, not homogenised properties, control cracking (Fig. 16). Creep strains after 10 h differ between families by more than an order of magnitude; the elastic graphite network carries load and reduces creep of the carbon-containing families.

9.2. Ranked BOF grades

For the five ranked grades (10 realizations each; creep at 5 MPa and 1550 °C) the calibration uncertainty is propagated exactly: $k_f(1 \pm \delta)$ maps damage at ΔT to damage at $\Delta T/(1 \pm \delta)$, with $\delta = 7.8\,\%$. After a 100 K shock MgO–C retains 0.87 (15 % C) and 0.92 (20 % C) of its stiffness and Al_2O_3–C 0.96, whereas MgO–ZrO_2 and magnesia retain 0.56 and 0.55 (Fig. 17, Table 12). The carbon grades also creep 5–19 times less (0.66–2.6 % against 12.4–12.8 % after 10 h). The microstructure model thus supports the outcome that the property-based ranking produced only when thermal-shock resistance was weighted strongly.

9.3. Surrogates

The descriptor network learns creep almost perfectly (cross-validated $R^2 = 0.98$ for creep rate and strain) and the onset reasonably ($R^2 = 0.72$), outperforming the random forest on every target and the Hasselman regression by a wide margin (Fig. 18). For E/E_0 at 100 K it reaches $R^2 = 0.57$: at the beginning of cracking the result depends on the positions of pores and flakes, which global descriptors do not contain. Leave-one-family-out errors are 1.3–4 times larger, so the network should be used within the sampled families. The dual U-Net predicts creep maps on held-out RVEs with pixel $R^2 = 0.98$–0.99 and crack maps with a mean intersection-over-union of 0.47 (pixel prior 0.20); it is exactly translation-equivariant, and the crack head is independent of the creep conditions by construction (Fig. 19). On the unseen grade realizations the surrogates reproduce the simulated ranking with Spearman coefficients of 0.90 (stiffness), 1.00 (creep) and 1.00 (creep maps), and the CNN cracked area lies within two standard deviations of the realization scatter for every grade.

10. Results of the three-dimensional models

10.1. Cohesive law in the RVE

The traction–separation histories of the first fully broken interface of each type follow the input law (Fig. 20): peak tractions of 35.9, 18.5 and 12.0 MPa against 38.4, 19.2 and 12.0 MPa, softening on the input lines, and an opening computed from the duplicated nodes identical to the element sepa-

Table 10. Comparison with published data of MgO–C bricks.

Quantity	Literature	This work	Remark
Young's modulus	5.5–13.5 GPa, three commercial MgO–C bricks, about 4–15 % graphite [36]	10.2–11.7 GPa (2-D grades), 13.2 ± 0.6 GPa (3-D RVE, 20 % C, 5 realizations)	within the measured range
Thermal expansion	8.6–11.3×10^{-6} K^{-1} at 300 °C [36]	11.0×10^{-6} (2-D), 11.1×10^{-6} K^{-1} (3-D)	model uses 20–1000 °C phase means
Fracture toughness K_{Ic}, MgO–20 % C, 298 K	1.26 ± 0.10 MPa$\sqrt{\text{m}}$ (19 tests, range 1.03–1.37) [37]	1.43 MPa$\sqrt{\text{m}}$ implied by E and G_F; crack start of the SENB at 0.93 kN	+14 %; G_F lies between the toughness- and wedge-splitting-based values
Fracture energy	240 J/m^2 at room temperature, 162–563 J/m^2 at 1100–1470 °C (wedge splitting) [29]	150 J/m^2 (brick models); 13–32 J/m^2 local (RVE phases)	brick value includes aggregate-scale bridging
Crack-initiation temperature difference	R = 171–330 K computed with the compressive strength [36]	66 K (index with f_t), 72 K (XFEM brick); first full interface failure at 100 K (3-D RVE)	compressive strength overestimates R 3–5×
Retained strength after one water quench, ΔT = 975 K, MgO–10 % C	47–87 % of the bending strength, depending on the binder [38]	gradual loss: 0.87–0.92 stiffness after 100 K (2-D), 12 % of elastic stress carried after 400 K under full constraint (3-D)	qualitative: no abrupt failure of MgO–C

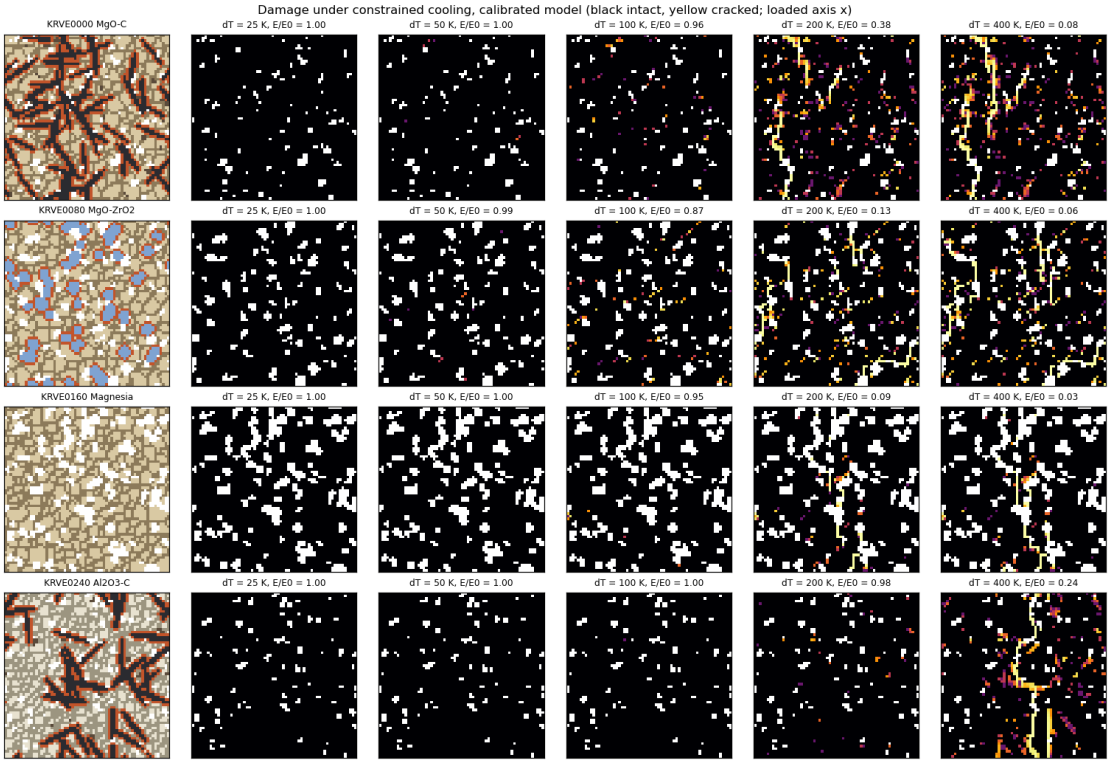


Figure 15. Damage evolution under constrained cooling for one RVE per family (calibrated model; black intact, yellow cracked, white pores; constrained axis x horizontal). Panel titles give ΔT and the retained stiffness E/E_0.

Table 11. Medians over 80 design RVEs per family. σ_t: uniaxial tensile strength.

	E/E_0 after			ΔT_c	E_0	σ_t
Family	50 K	100 K	200 K	[K]	[GPa]	[MPa]
MgO–C	0.984	0.896	0.227	26.2	10.6	13.2
MgO–ZrO_2	0.960	0.727	0.077	32.0	12.7	12.4
Magnesia	0.972	0.749	0.055	34.2	12.1	12.9
Al_2O_3–C	0.999	0.984	0.879	48.3	10.8	16.0

ration. Graphite and binder interfaces start to damage first (at 0.076 % strain), but the binder interface breaks last because its softening branch is the longest ($2G_c/t^0 = 1.07$ µm). The small rising traction after failure is the 1 % residual stiffness of fully damaged interfaces.

Table 12. Ranked grades (mean ± standard deviation of 10 realizations; creep 10 h at 5 MPa and 1550 °C).

Grade	E_0 [GPa]	ΔT_c [K]	E/E_0, 100 K	Creep [%]
MgO–C 15 % C	11.7 ± 0.5	25 ± 5	0.87 ± 0.02	2.58 ± 0.11
MgO–C 20 % C	10.2 ± 0.8	29 ± 9	0.92 ± 0.03	2.04 ± 0.12
MgO–ZrO_2	15.1 ± 1.1	27 ± 6	0.56 ± 0.12	12.8 ± 0.9
Magnesia 97	13.2 ± 0.7	30 ± 5	0.55 ± 0.16	12.4 ± 0.7
Al_2O_3–C	13.4 ± 0.4	34 ± 10	0.96 ± 0.01	0.66 ± 0.02

10.2. Tension (A1)

The response is nearly linear with $E_0 = 13.3$ GPa up to about 0.14 % strain and reaches a peak of 20.0 MPa at 0.168 % (Fig. 21). Graphite interfaces fail first (from 0.145 %), grain boundaries from 0.175 %, and binder interfaces hardly (0.5 % broken at the end, against 12.3 % of the graphite interfaces

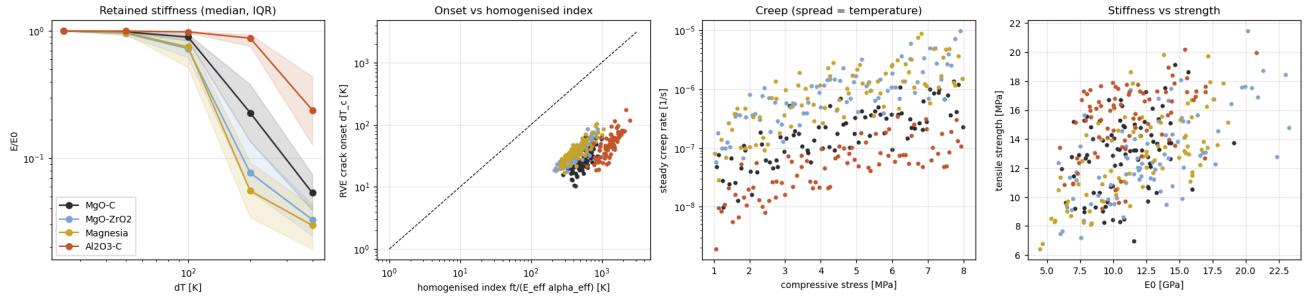


Figure 16. 320 design RVEs. From left: retained stiffness versus ΔT (median and interquartile range); crack onset versus the homogenised index $f_t/(E_{eff}\alpha_{eff})$ (dashed: equality); steady creep rate versus stress (spread: temperature); tensile strength versus E_0. Colours: MgO–C black, MgO–ZrO_2 blue, magnesia yellow, Al_2O_3–C red.

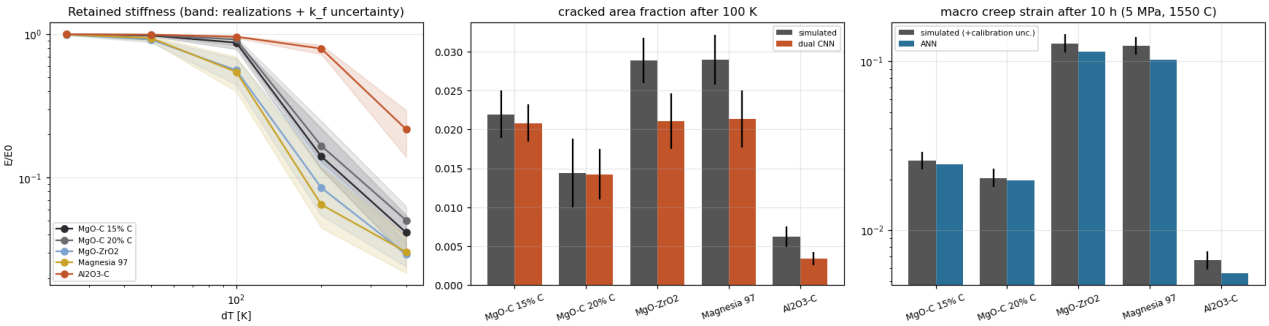


Figure 17. Ranked grades. Left: retained stiffness with realization scatter and k_f uncertainty. Middle: cracked area after 100 K, simulation and CNN. Right: creep strain after 10 h (5 MPa, 1550 °C), simulation and ANN.

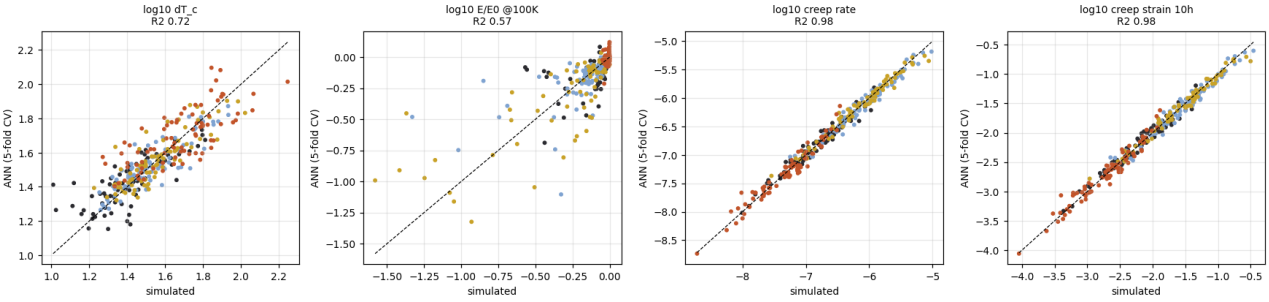


Figure 18. Descriptor network, 5-fold cross-validation on the 320 design RVEs: crack onset, retained stiffness at 100 K, creep rate and creep strain after 10 h (colours as in Fig. 16).

and 3.5 % of the grain boundaries). Only 25 interfaces are broken at the peak; at the end 1054 broken interfaces form one rough band normal to the load (Fig. 22): the crack follows the graphite flakes and links them through MgO grain boundaries. The dissipated energy (2.85 µJ) divided by the broken area (≈0.46 mm^2) gives about 6 J/m^2, between the graphite-interface and grain-boundary toughness. The RVE strength exceeds the brick tensile strength because a 0.5 mm cube contains no millimetre flaws.

10.3. Thermal shock (A2)

Up to 100 K both in-plane stresses follow the independent elastic solution within 3 %, with $\sigma_{zz} = 1.5\,\sigma_{xx}$ because the flakes lie across x (Fig. 23). The independent solution predicts the first damage initiation at 37 K (binder interfaces; 40.5 ± 3.0 K over five realizations, Table 8); the first interfaces fail fully at 100 K (grain boundaries), followed by graphite interfaces (140 K) and binder interfaces (180 K). σ_{zz} peaks at 24 MPa near 120 K and σ_{xx} at 19 MPa near 140 K. After 400 K, 20 % of all interfaces are broken (grain boundaries 20 %, graphite interfaces 24 %, binder 3 %). The broken interfaces first form isolated walls along the flakes and grain boundaries (1160 at 200 K) and then connect into a three-dimensional network (2780 at 400 K; Figs. 24 and 25). Nevertheless the hot face still carries 27 % of the elastic stress at 200 K and 12 % at 400 K, against 10 and 3 % for the 2-D model of the same grade: load paths around the cracks remain in three dimensions, whereas a single line of broken pixels separates a 2-D section. Locally, the misfit between graphite and MgO keeps the phases loaded: von Mises stresses of 30–50 MPa in graphite and 20–40 MPa in MgO at 400 K, with peaks of 141 MPa along a line through the RVE, while the macroscopic stress is 8 MPa (Fig. 26).

10.4. Creep (C1–C3)

After 10 h at 5 MPa the macroscopic creep strain is 0.16, 0.32 and 0.58 % at 1350, 1450 and 1550 °C (Fig. 27). At 1550 °C the 2-D model of the same grade gives 2.04 %, i.e. 3.5 times more, although both models are calibrated on the same stiffness and creep data. The reason is load transfer: the

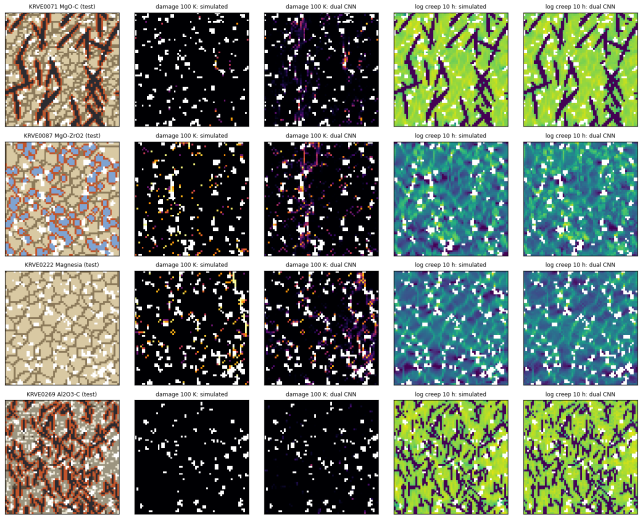


Figure 19. Dual-head U-Net on held-out RVEs: microstructure, simulated and predicted damage after 100 K, simulated and predicted log creep strain after 10 h.

mean von Mises stress in the graphite rises from 7 to 21 MPa while the MgO relaxes from 7 to 2.5 MPa, and the connected graphite skeleton, which does not creep, carries the load (Fig. 28). As a consequence the creep rate decreases with time ($\varepsilon \propto t^{0.6}$) although the constitutive law has no primary stage, and the macroscopic temperature sensitivity is lower than that of the input law (a factor of 2 per 100 K, apparent activation energy ≈160 kJ/mol, against 281 kJ/mol). Local creep strains are about three times the macroscopic value (grain boundaries 1.9 %, MgO 1.6 %, interfaces 1.1 % at 1550 °C).

10.5. Hot-face slice (B2)

The first crack forms at a surface temperature drop of 72 K. Cracks grow rapidly between 72 and 150 K and stop at about 250 K with 206 cut elements, limited by the cooled depth of about 11 mm (Figs. 29 and 30). They run from the hot face into the brick, perpendicular to it. Only the two outer enrichment strips crack; the middle strip remains intact because the two cracks relieve the stress between them, so the cracks space themselves out. This is the precursor of spalling: a longer cold exposure deepens the cooled layer ($\propto \sqrt{\kappa t}$) and with it the cracks.

10.6. Bending bar (B1)

In three-point bending the crack starts at mid-span on the tensile face when the bottom-fibre stress reaches f_t and grows through the depth (Fig. 32). With $2.5 \times 1.56 \times 2.5$ mm elements the peak force is 1.29 kN at a load-point deflection of 0.147 mm, i.e. a modulus of rupture $\mathrm{MOR} = 3FS/(2BH^2) = 15.5$ MPa and $\mathrm{MOR}/f_t = 1.35$, the stress-gradient effect expected for quasi-brittle bending of a 25 mm bar. A coarser mesh with 5 mm elements gives 1.41 kN (17.0 MPa); halving the element size lowers the strength by 9 %, the expected trend for XFEM cracks whose initiation stress is evaluated over a smaller volume. After the peak the force drops abruptly to about 20 % of the peak, the brittle snap-back typical of displacement-controlled bending of a quasi-brittle bar, and then decreases step-wise as the crack front advances row by row through the enriched elements (Fig. 31). The crack-mouth opening is 8 µm at the peak and 0.21 mm at the end. Energy balance at the end of the test, per ligament area BH: external work 203 J/m^2, of which 125 J/m^2 is dissipated by the crack, 37 J/m^2 by the numerical damage stabilisation and 33 J/m^2 remains as elastic energy. Since the bar still carries 10 % of the peak force, the ligament is not yet fully separated, and the crack dissipation is consistent with the input $G_F = 150$ J/m^2 for a crack that has opened over about 80 % of the ligament. Along the bottom fibre the stress is highest at mid-span before cracking and drops to zero where the crack opens, while two maxima move outwards. The modulus of rupture of 15.5 MPa lies in the range measured for MgO–C bricks with 7–15 wt% graphite (13–16 MPa [27]).

10.7. Notched beam (D1)

Along the crack front K_I is highest at mid-thickness, 48.3 MPa$\sqrt{\text{mm}}$ at $P = 1$ kN (0.99 of the ASTM E399 value $K = PS/(BW^{1.5})f(a/W) = 48.7$ MPa$\sqrt{\text{mm}}$, $f(0.3) = 1.521$), and decreases towards the free surfaces, the classical three-dimensional constraint effect (Fig. 33). The contours agree within 0.4 %, a coarser mesh gives 0.97 of the ASTM value, and an independent compliance calculation gives 48.9 MPa$\sqrt{\text{mm}}$. With $J = K^2(1-\nu^2)/E = 0.170$ N/mm at 1 kN, a crack of 7.5 mm starts to grow at $P = 1\,\text{kN}\sqrt{G_F/J} \approx 935$ N, first in the interior of the brick where K is highest.

10.8. Three-dimensional check of the two-dimensional result: what changes and why

Table 13 compares the 2-D results of the MgO–C 20 % C grade with the 3-D models of the same grade. The qualitative conclusions of the 2-D model are confirmed: cracking starts at weak interfaces and grain boundaries at a few tens of kelvin, cracks run into the depth normal to the held direction, damage accumulates gradually, and graphite carries load and limits creep. The quantitative differences have three physical causes.

1. *Connectivity.* In a plane-stress section every flake, boundary and crack is a prism through the thickness. One line of broken pixels separates a 2-D RVE, whereas in 3-D cracks must link over a surface and the stiff phases find load paths around them. This is why the 3-D hot face still carries 27 and 12 % of the elastic stress after 200 and 400 K, against 10 and 3 % in 2-D.
2. *Graphite skeleton.* In 3-D the graphite flakes form connected networks (33 in the RVE) that do not creep; they take over the load released by the creeping MgO. In 2-D the flakes are isolated and the MgO carries the full load. Both models are calibrated on the same stiffness and creep data, so the 3.5-fold lower creep in 3-D is a microstructural effect, not a calibration effect.
3. *Constraint.* The 3-D hot face is held in both in-face directions, which raises the thermal stress by $1/(1-\nu) \approx 1.2$ relative to the 2-D model and makes the flake orientation matter ($\sigma_{zz} = 1.5\sigma_{xx}$). This shifts damage to the grain boundaries, which are loaded by both stress components, whereas in tension and in 2-D the graphite interfaces dominate.

The 2-D model is therefore conservative for the carbon grades, and the grade ranking derived from it is unchanged.

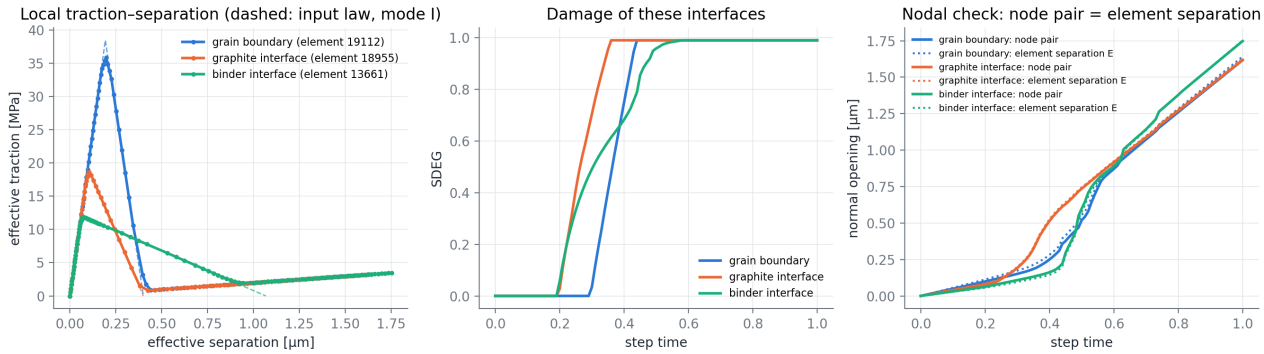


Figure 20. Cohesive interfaces in the 3-D RVE (A1). Left: traction–separation of the first fully broken grain-boundary, graphite and binder interface (dashed: input laws). Middle: damage SDEG of these interfaces. Right: opening from the duplicated node pair (solid) and element separation (dotted).

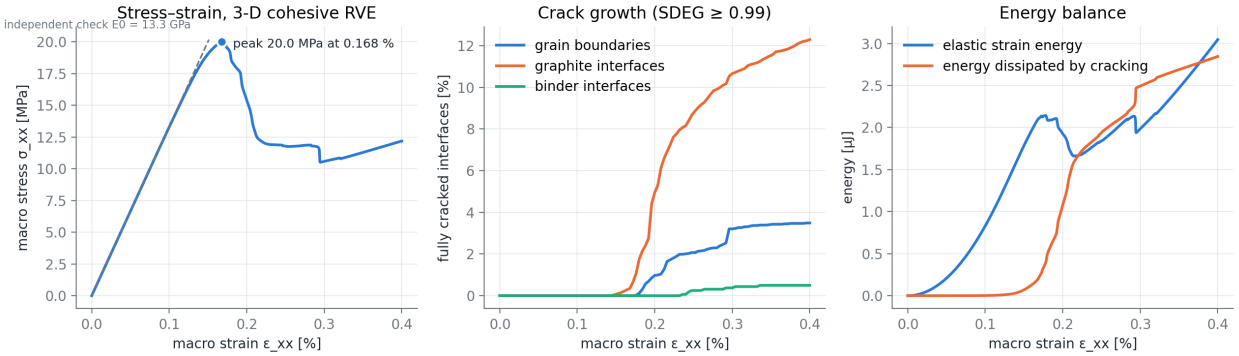


Figure 21. Tension of the 3-D RVE (A1). Left: macroscopic stress–strain curve with the independent elastic check. Middle: fully broken interfaces per type (SDEG ≥ 0.99). Right: elastic strain energy and energy dissipated by cracking.

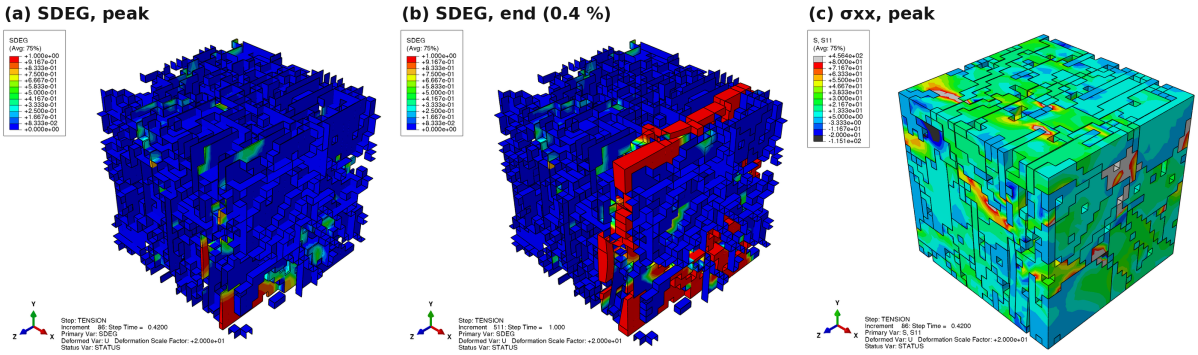


Figure 22. Abaqus results of A1 (deformation $\times 20$, interfaces only in a and b). (a) Damage SDEG at the peak stress, (b) at 0.4 % strain: the broken band normal to x; (c) σ_{xx} at the peak in the bulk phases.

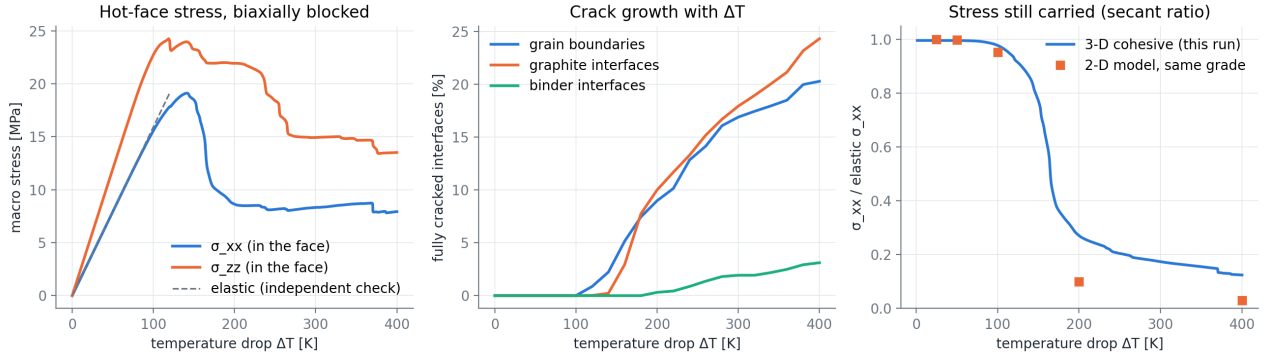


Figure 23. Thermal shock of the 3-D RVE (A2). Left: in-plane stresses σ_{xx}, σ_{zz} with the independent elastic solution. Middle: fully broken interfaces per type. Right: secant ratio $\bar{\sigma}_{xx}/\sigma^{el}_{xx}$, 3-D cohesive model and 2-D model of the same grade.

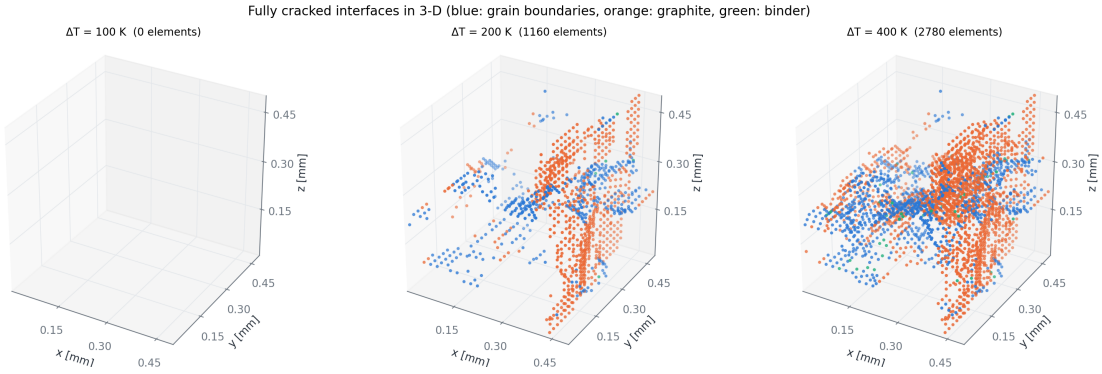


Figure 24. Fully broken interfaces after 100, 200 and 400 K of cooling (blue: grain boundaries, orange: graphite interfaces, green: binder).

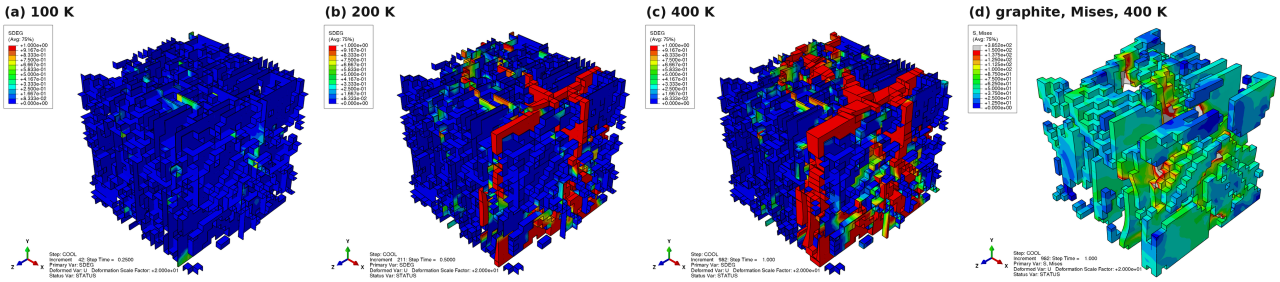


Figure 25. Abaqus results of A2: interface damage SDEG after (a) 100 K, (b) 200 K and (c) 400 K (deformation ×20); (d) von Mises stress in the graphite after 400 K (scale 0–150 MPa; grey above).

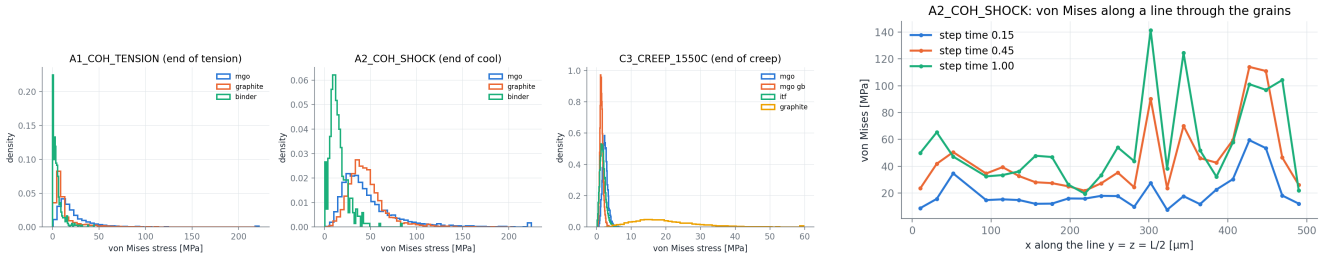


Figure 26. Local stresses. Left: von Mises distributions per phase at the end of tension, cooling and creep. Right: von Mises along the line $y = z = L/2$ after 60, 180 and 400 K of cooling. Single elements above 200 MPa are voxel corners next to open cracks.

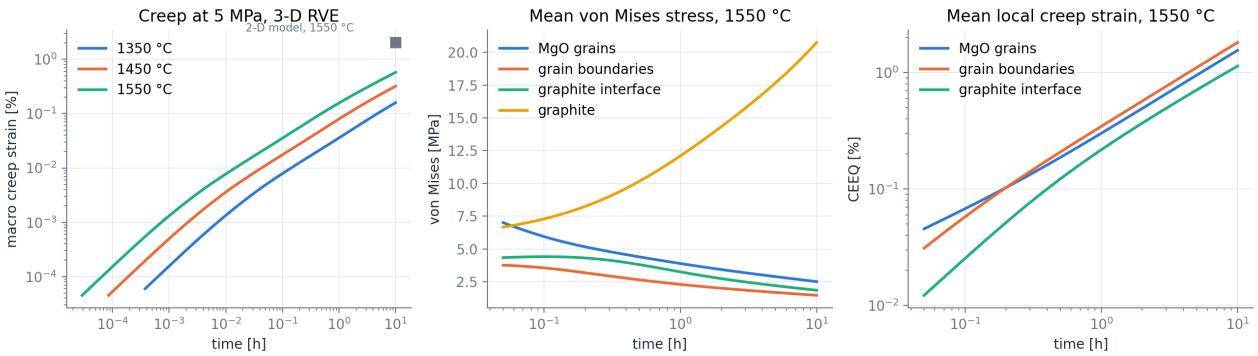


Figure 27. Creep of the 3-D RVE (5 MPa, 10 h). Left: macroscopic creep strain at three temperatures (square: 2-D model, 1550 °C). Middle: mean von Mises stress per phase at 1550 °C. Right: mean local creep strain per phase.

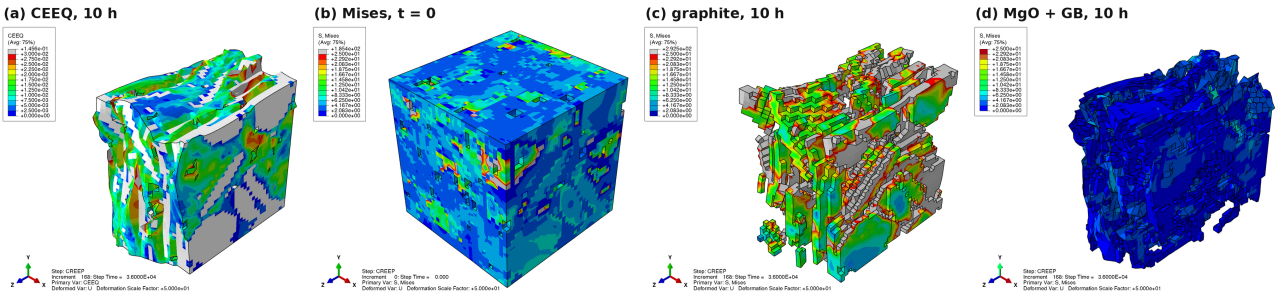


Figure 28. Abaqus results of C3 (1550 °C; deformation ×50). (a) Equivalent creep strain after 10 h (scale 0–3 %; uncoloured: non-creeping graphite and binder). (b) von Mises stress at the start of creep. (c) Graphite and (d) MgO with grain boundaries after 10 h (same scale 0–25 MPa).

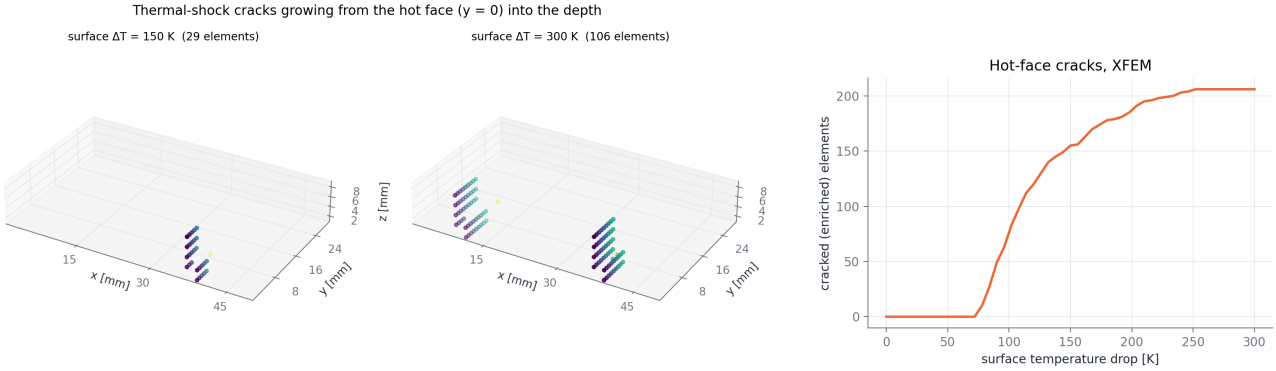


Figure 29. Hot-face slice B2. Left: cracked XFEM elements after surface drops of 150 and 300 K. Right: number of cracked (enriched) elements versus the surface temperature drop.

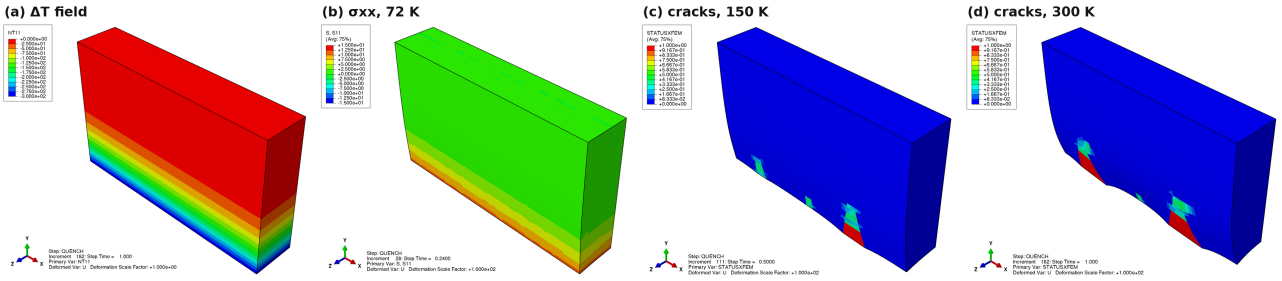


Figure 30. Abaqus results of B2 (hot face at the bottom; deformation ×100 in b–d). (a) Temperature change at the end of cooling. (b) σ_{xx} just before the first crack (72 K). (c, d) XFEM crack status after 150 and 300 K (red: fully cracked).

Table 13. MgO–C 20 % C: two-dimensional model versus the three-dimensional check.

Quantity	2-D model (10 realizations)	3-D check (Abaqus)	Cause of the difference
Crack description	smeared saw-tooth damage, SLA	cohesive interfaces (RVE), XFEM (brick)	independent formulation and solver
Hot-face constraint	x held, plane stress	x and z held	stress $\times 1/(1-\nu) \approx 1.2$ in 3-D
Stiffness E_0	10.2 ± 0.8 GPa	13.2 ± 0.6 GPa (cohesive RVE, 5 seeds), 12.0 GPa (creep RVE)	3-D load paths; creep RVE re-calibrated in 3-D
First cracking	29 ± 9 K (first event)	40 ± 3 K first damage (5 seeds), 100 K first broken interface	same order; full opening needs linked damage
First phase to break	weak interfaces, grain boundaries, pores	graphite interfaces (tension); grain boundaries (biaxial cooling)	biaxial stress loads boundaries of all orientations
Crack pattern	through-crack normal to x at 200–400 K	walls into the depth, then a 3-D network (20 % of interfaces at 400 K)	cracks must link over a surface
Stress carried, 200 / 400 K	10 / 3 % of elastic	27 / 12 % of elastic	load paths around cracks in 3-D
Creep, 10 h, 5 MPa, 1550 °C	2.04 %	0.58 %	connected graphite skeleton carries load
Creep kinetics	constituent law ($Q = 274$ kJ/mol)	$\varepsilon \propto t^{0.6}$, apparent $Q \approx 160$ kJ/mol	load transfer to graphite
Grade ranking	MgO–C most damage tolerant	confirmed	–

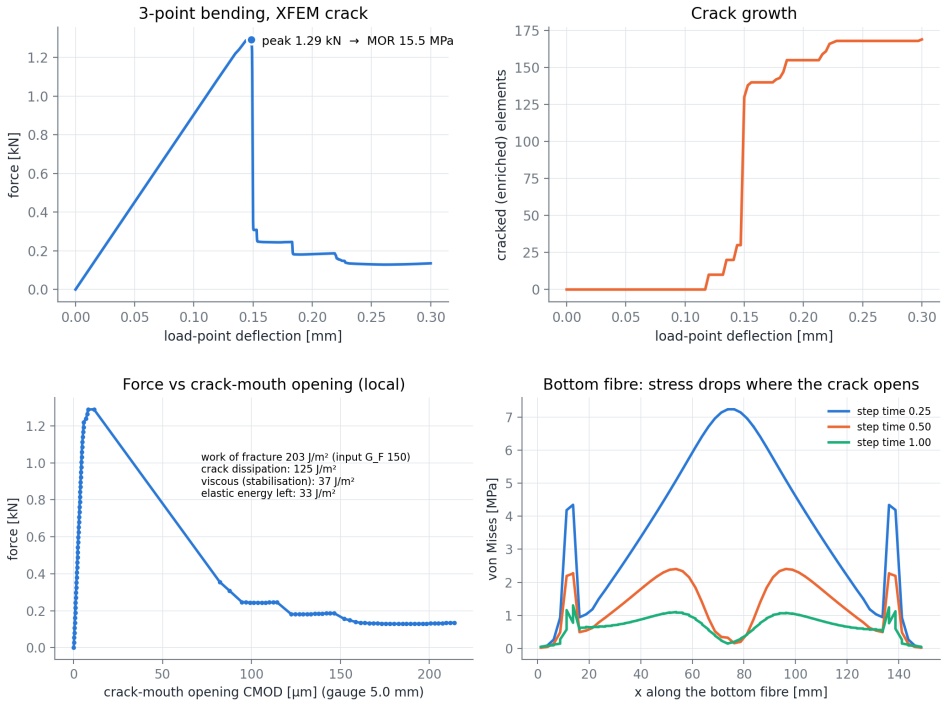


Figure 31. Bending bar B1 (2.5 mm mesh). Top: force versus load-point deflection (peak 1.29 kN, MOR 15.5 MPa) and number of cracked XFEM elements. Bottom: force versus crack-mouth opening (gauge 5 mm) with the energy balance, and von Mises stress along the bottom fibre at step times 0.25, 0.5 and 1.

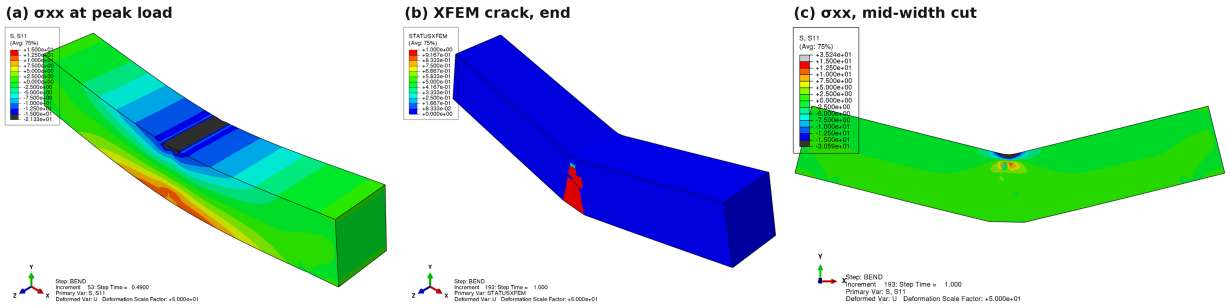


Figure 32. Abaqus results of B1 (2.5 mm mesh; deformation ×50). (a) σ_{xx} at the peak load (scale −15 to 15 MPa). (b) XFEM crack at mid-span at the end. (c) σ_{xx} on a cut at mid-width: compression under the load, crack tip below it.

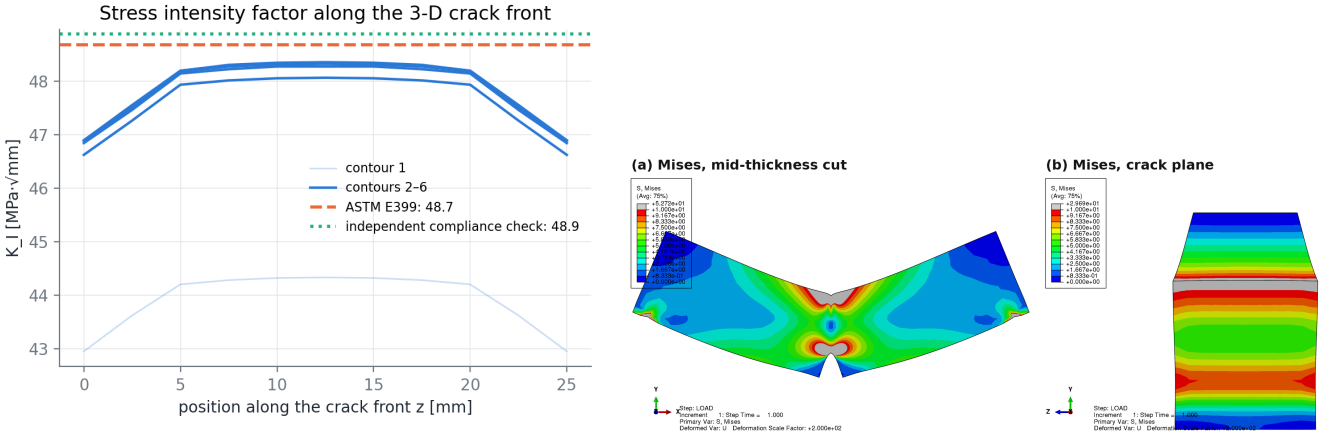


Figure 33. Notched beam D1. Left: K_I along the crack front (contour 1 and contours 2–6; dashed: ASTM E399; dotted: compliance calculation). Right: von Mises stress (scale 0–10 MPa, deformation ×200) on a cut at mid-thickness and on the crack plane, showing the loaded crack front across the thickness.

11. Discussion

11.1. Crack initiation versus damage resistance

The results separate two properties that index-based selection mixes: resistance to crack initiation (R) and resistance to damage accumulation (R'''') [2]. Graphite flakes and weak interfaces concentrate stress and nucleate many short cracks early, so MgO–C has the lowest onset; but the compliant, low-expansion graphite network relieves the misfit stress and arrests these cracks, so its retained stiffness after moderate shocks is the highest of the MgO-based families. Dense oxide grades fail later but abruptly. The 3-D cohesive model shows the same sequence on the real interface network, now with the hot face held in both in-face directions, and quantifies the damage tolerance: after 400 K the cracked hot face still carries 12 % of the elastic stress. Because this behaviour emerges from the microstructure and not from fitted damage parameters, the model provides a mechanistic basis for the industrial choice of MgO–C and for a high weight of thermal-shock damage in multi-criteria selection.

11.2. What the three-dimensional check adds

The 3-D models were designed as a check of the 2-D result for one grade, and they answer three questions that the 2-D model cannot: whether the conclusions survive a different crack formulation and solver (they do), how large the plane-stress bias is (3–4 times in residual stress, 3.5 times in creep, both on the conservative side for carbon grades), and which microstructural feature causes it (the connectivity of cracks and of the graphite skeleton, Section 10.8). Because the 2-D model underestimates the in-face constraint by $1/(1-\nu) \approx 1.2$ while overestimating crack connectivity, the two effects partly compensate for crack initiation, which explains why the onsets of both models are of the same order.

11.3. Two scales

The brick-scale models use the stiffness of the RVE and brick-level fracture parameters, and they reproduce three independent brick-scale facts: the critical temperature difference of a held face, a bending strength in the measured range, and stress intensity factors consistent with standard fracture mechanics and with published toughness. Their fracture parameters are not yet derived from the RVE; a submodelling approach that embeds a resolved microstructure window at the notch tip or in the cooled layer of the brick model is the natural next step.

11.4. Limitations

(i) Measured retained properties after quenching of the investigated grades were not available; the thermal-shock predictions are verified and compared with published data (Section 8.2) but not validated quantitatively. (ii) Plane stress leaves the second in-face direction free; the 3-D RVE removes this restriction for one grade. (iii) A 1 mm RVE contains grains, flakes and pores but not millimetre aggregates, which is the likely reason for local fracture energies (13–32 N/m) far below brick-scale values (240 N/m [29]). Statistical RVE-size criteria [41] and a two-scale model are planned. (iv) One boundary stiffness factor cannot be separated into grain-boundary and interface contributions with the present data. (v) One power law cannot follow the temperature-dependent stress exponent of the magnesia data; the creep model is therefore restricted to $\sigma \geq 2.5$ MPa, and a linear-plus-power law calibrated on low-stress data would extend it. The magnesia modulus cannot be validated until the porosity of that brick is known. (vi) Beyond 100 K the residual stress depends on the unloading description (secant unloading in SLA, plastic crack opening and viscosity in concrete damaged plasticity). (vii) The 3-D damage evolution and creep were computed for one microstructure realization; five realizations give a scatter of 0.4–7 % in stiffness, hot-face stresses and damage onset (Table 8). Cracked interfaces retain 1 % stiffness for numerical stability. (viii) Oxidation of carbon, slag infiltration and temperature-dependent elastic and fracture properties are not modelled.

12. Conclusions

1. A data-driven ranking based on ML phase properties and TOPSIS shows that the choice of BOF lining material depends on the weight of thermal-shock damage, a quantity that index-based selection cannot resolve.
2. Periodic Kanapy RVEs sampled by a Latin-hypercube design, combined with an energy-consistent sequentially linear analysis and Gauss-point Norton creep, provide a robust and inexpensive simulator (about 1–2 CPU minutes per RVE and load set).
3. Five parameters calibrated on one MgO–C brick and one creep series predict the graphite effect on stiffness and strength of a second brick within -16 and $-6\,\%$ and magnesia creep at 1500 °C within 0.22 dex for $\sigma \geq 1.5$ MPa; an independent Abaqus solution of 60 RVEs agrees in stiffness, creep (0.5 %) and post-shock stiffness up to 100 K (0.016).
4. MgO–C initiates cracks first but retains 0.87–0.92 of its stiffness after 100 K, against 0.55–0.56 for MgO–ZrO_2 and magnesia, and creeps 5–19 times less.
5. Three-dimensional cohesive and creep RVEs of MgO–C 20 % C confirm the mechanism on the real interface network: a distributed microcrack network that keeps 12 % of the elastic hot-face stress after 400 K, and creep reduced 3.5-fold by the connected graphite skeleton.
6. Brick-scale XFEM and fracture-mechanics models give a hot-face crack onset of 72 K, a modulus of rupture of 15.5 MPa (measured 13–16 MPa) and stress intensity factors within 1 % of ASTM E399; the implied toughness agrees with published data within 14 %.
7. Descriptor and periodic image surrogates reproduce the simulated ranking of the grades in milliseconds and are suitable for screening within the sampled families.

Use of artificial intelligence

Machine learning as part of the method. ML is used at three places: (1) gradient-boosting and transfer-learning models trained on DFT and experimental data provide phase properties for the ranking (Section 3); (2) a descriptor network and (3) a periodic dual-head U-Net, trained exclusively on the simulations of this work, act as surrogates of the microstructure simulator (Sections 4.6 and 9.3). The surrogates reproduce the simulator, not experiments; they were evaluated by cross-validation with splits by whole microstructure and on unseen realizations of the ranked grades.

Generative AI assistance. During the preparation of this work the authors used Claude (Anthropic), a large language model, to assist with writing and debugging code (microstructure generation, solvers, Abaqus input-deck generators, post-processing scripts), generating figures and drafting the text. All results were produced by the authors' simulations; every code path was checked by the exact tests, by the independent Abaqus solutions and by the input-deck validators described in Section 6. The authors reviewed and edited all content and take full responsibility for the publication.

CRediT authorship contribution statement. P. Sreenath: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **A. Hartmaier:** Supervision, Methodology (Kanapy), Writing – review and editing.

Declaration of competing interest. The authors declare no competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data availability. Microstructures, simulation results, Abaqus input decks, extraction scripts and the notebooks that reproduce all figures will be made available in a public repository upon publication.

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